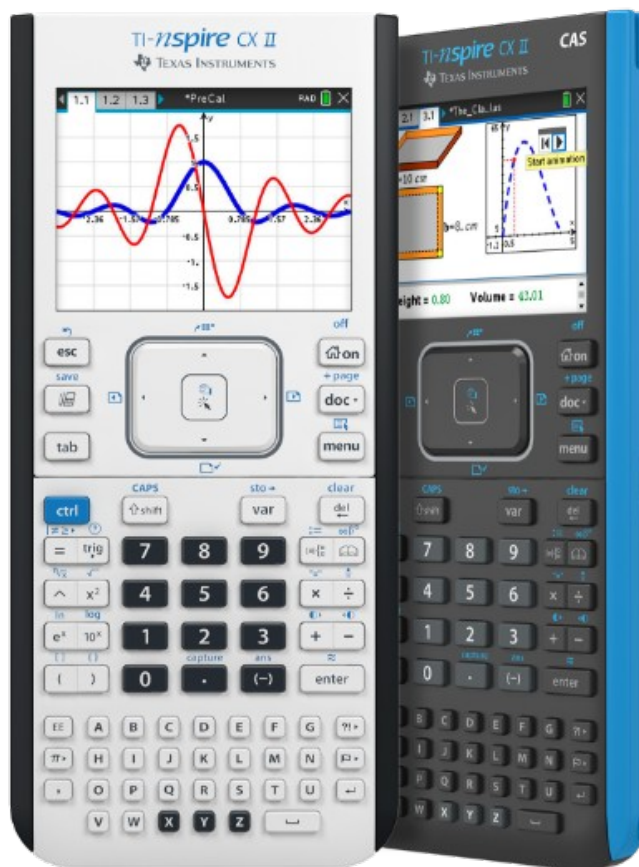


# StepCAS / nps

## *The illustrated beginner's handbook*

Your calculator, your first problem and the steps behind the answer.



## *TI-Nspire CX II • Ki V4 • Current development edition*

9 September 2026

Prepared for Ian's personal study and reference.

Includes step-by-step lessons, annotated pictures, practice problems, troubleshooting and the current menu reference.

Cover photograph: Texas Instruments. This handbook describes the inspected project source and labels historical screenshots and unverified device behavior.

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## 01 / LEARN &amp; USE

# Start here

## *What this manual covers and how to use it*

This is a practical guide to StepCAS, the nps application presented through Ki V4 on the TI-Nspire CX II. It starts with the keys and follows the ordinary tasks of entering mathematics, reading explanations and saving work. You do not need to know Lua, C++ or how the calculator was built.

Read Your first session if the app is already installed. Use Opening StepCAS if you are at the Home screen. The command reference near the end is for looking things up after you have tried the tutorials.

| Name you may see          | What it means                                                        |
|---------------------------|----------------------------------------------------------------------|
| TI-Nspire CX II           | The physical calculator and its built-in operating system.           |
| nps / StepCAS             | Our application and its mathematical walkthrough engine.             |
| Ki V4 / nps_v4            | The document that displays the entry line, history and walkthroughs. |
| Giac / KhiCAS             | The algebra engine used for ordinary calculations and some checks.   |
| ndl / older ndless labels | The runtime that allows the native module to run.                    |

### Edition and evidence

Prepared 9 September 2026 from the current working source. This is an unreleased development build, not a promise that every feature is on your installed package. The pictures use retained emulator captures and TI product photographs. Annotated plates are teaching illustrations. They are not new photos of your handheld or proof of a fresh device test.

Expected answers in the lessons are mathematical targets. Equivalent algebraic forms are acceptable. A screenshot can show older wording while the nearby instructions describe the current source. The manual covers everyday calculator use and the current StepCAS interface. Advanced TI applications and the much larger Giac command library have their own linked references.

## 02 / LEARN &amp; USE

# Your first session

*Get one result, open its explanation and return to the entry line*

1. Turn on the calculator with Home / On. Choose Browse under Documents and open the installed StepCAS document, usually named nps\_v4.
2. Wait for the application to finish loading. A Giac OK message identifies the algebra engine. Read any dependency or integrity error before proceeding.
3. Select the entry box at the bottom. Type  $!s \ 2*x+3=11$  and press Enter. The exclamation mark is available from the punctuation palette.
4. Look for the answer  $x = 4$ . In the step list, use Up and Down to select a step. Press Enter to open its explanation.
5. Press Shift+Tab to switch between Standard and Beginner detail. Beginner gives more explanation. It does not change the answer.
6. Press Escape once to return to the list and again to return to the entry screen. Save the document with Ctrl+S.

`!s 2*x+3=11`

## What you should understand

Subtracting 3 from both sides gives  $2*x = 8$ . Dividing both sides by 2 gives  $x = 4$ . Substituting 4 into the original left side gives 11.

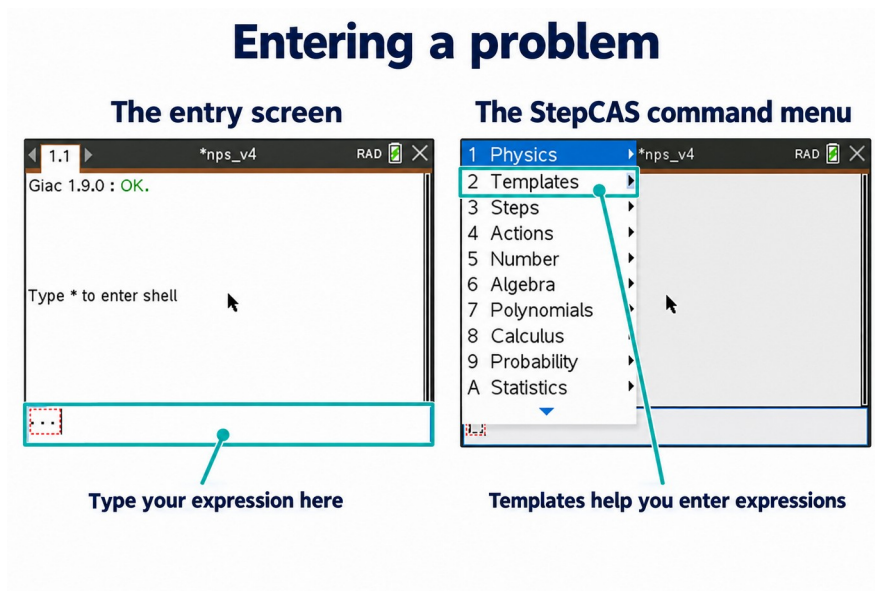


Figure 1. Annotated retained emulator captures. The bottom box accepts input. The old “Type \* to enter shell” message opens an additional native shell, which is not needed for this lesson.

## 03 / LEARN &amp; USE

# Meet the calculator

*Find the physical controls before learning shortcuts*

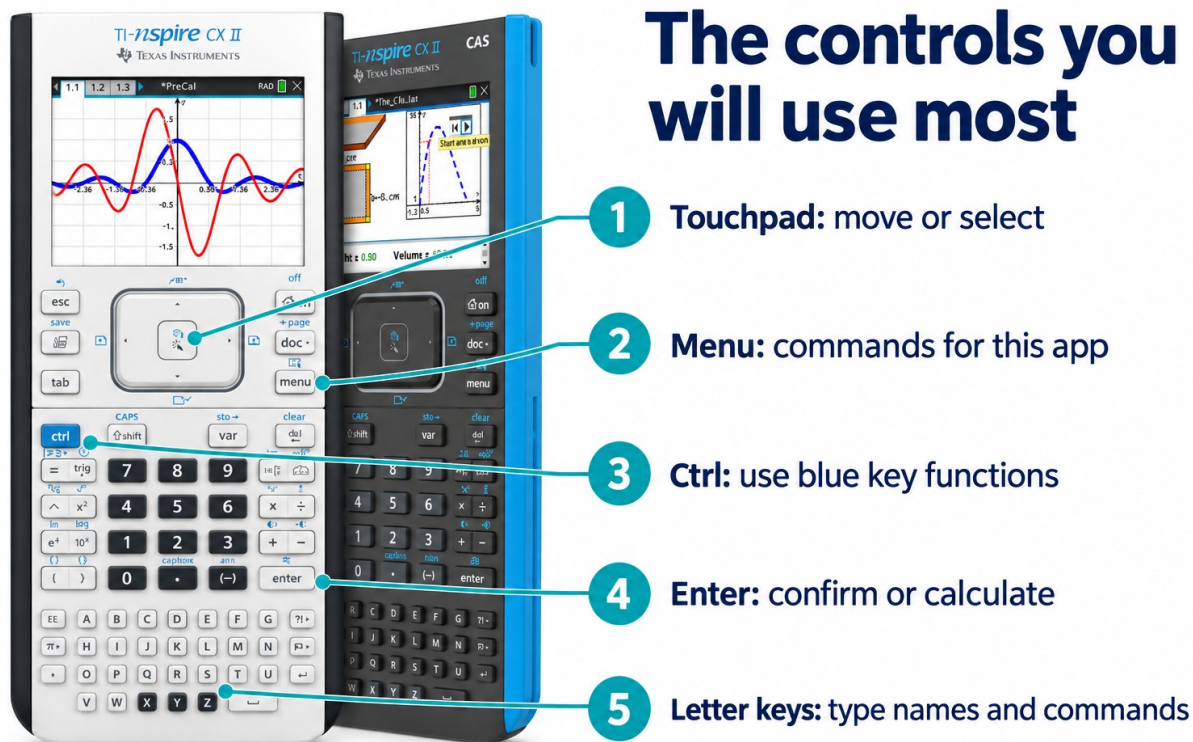


Figure 2. Annotated TI product photo. The white calculator is the CX II. The darker model behind it is a CAS model. Their built-in symbolic capabilities differ.

Press the outer edges of the touchpad for Up, Down, Left and Right. Sliding a finger moves the pointer. The center click selects an item. For a first lesson, the directional edges and Enter are easier to control than pointer movement.

Ctrl is blue. Hold it while pressing another key to use that key's blue function. Shift changes the next character or activates a shifted shortcut. A shortcut written Ctrl+S means hold Ctrl, press S, then release.

## You are using two layers

Home and Doc belong to the calculator system. StepCAS supplies its own entry screen, menu and walkthrough controls. A key can do different things in those places. The screen's footer is the best immediate reminder.

Source: TI Handheld Keys and Keyboard Shortcuts [T1, T2].

## 04 / LEARN &amp; USE

# The everyday keys

*A small set is enough for most work*

| Key              | Use                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------|
| Home / On        | Power on or return to the system Home screen.                                                     |
| Ctrl + Home / On | Power off. Save the current document first.                                                       |
| Doc              | Open the system document menu.                                                                    |
| Menu             | Open commands for the current application.                                                        |
| Enter            | Confirm a choice. In the StepCAS entry box, submit the expression.                                |
| Escape           | Dismiss a menu or go back in the current view.                                                    |
| Tab              | Move between fields. Inside a StepCAS hint walkthrough, reveal more work.                         |
| Shift + Tab      | Move backward through fields. In a StepCAS walkthrough, change explanation detail.                |
| Del              | Delete the character before the cursor in an editable field.                                      |
| Ctrl + Del       | Clear the focused input. When history is focused, StepCAS can delete that entry.                  |
| Ctrl + S         | Save the current document.                                                                        |
| Ctrl + C / V     | Copy or paste when an editable selection has focus.                                               |
| Ctrl + Z         | Request Undo in an editor that supports it. It does not undo a completed mathematical derivation. |

## Escape is not a reset button

In step details, Escape returns to the list. From the step list it returns to the entry screen. Use Home to reach the calculator Home screen. Do not use Ctrl+Escape as a universal “back” shortcut. Its behavior is context dependent.

For punctuation such as !, quotes and a colon, open the punctuation palette. For square brackets, use the blue bracket function with Ctrl and the parenthesis key. Move the cursor out of a fraction or exponent before typing the next argument.

Sources: [T1, T2] and current StepCAS event handling.

## 05 / LEARN &amp; USE

# Opening StepCAS

*Start from Home and choose the document*

## Finding StepCAS

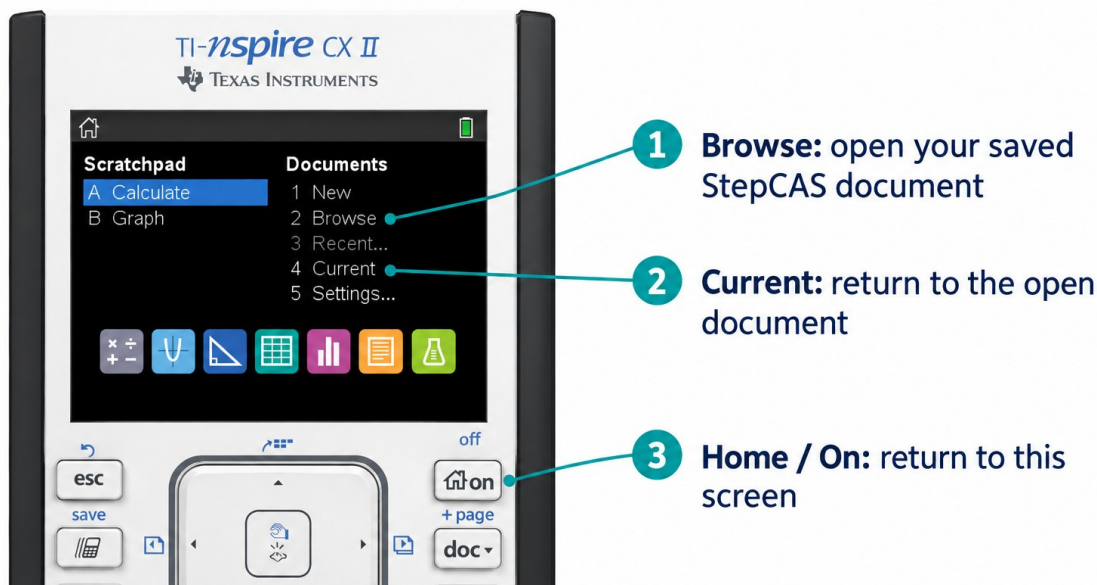


Figure 3. Annotated TI photograph showing Browse, Current and Home / On. This is the system Home screen, not the StepCAS menu.

1. Press Home. Under Documents, choose Browse. Use Up and Down to highlight a folder and Enter to open it.
2. Open the folder containing the installed StepCAS document. The current project layout uses an nps folder. Earlier installations used other folders.
3. Select nps\_v4 and press Enter. The calculator may hide the .tns extension. Select the document, not the nps\_nspire.luax module.
4. If StepCAS is already open, use Current from Home. Recent can also help you find the document.

### If the file is missing

Do not rename an installer or module to make it look like the document. Use the installation checklist in this manual. The app needs a matching document, native module and integrity sidecars.

Sources: TI Home and document navigation [T3, T4].

## 06 / LEARN &amp; USE

# Installation and version checks

## *Recognize the package before changing anything*

This manual assumes a working runtime installation. Setting up or repairing ndl depends on the handheld model and operating-system version. A generic OS update or installer sequence is not safe to assume for this development setup.

| Current project layout           | Purpose                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------|
| /nps/nps_v4.tns                  | The document you open. Keep its accompanying sidecar.                             |
| /nps/runtime/nps_nspire.luax.tns | The unified native module. Keep its accompanying sidecar.                         |
| /ndl                             | Runtime resources and installation files. These are not ordinary worksheet files. |

Older captures use ndless folder names. Those pictures help identify the browser, but they are not an installation manifest for the current package.

1. On the calculator, find Settings and Handheld Status to record the model and OS version.
2. Open StepCAS and type !m on its own line. Record the displayed package and backend information.
3. If the app reports a missing export, incompatible module or failed integrity check, preserve the exact message. Replace the package only with a matching prepared set.
4. Keep a backup of your documents before transferring or replacing application files.

```
! m
```

### Do not repair a mismatch by trial and error

Do not delete runtime folders, reset storage, install arbitrary firmware or mix old and new sidecars. A desktop build does not establish that a package loaded correctly on the handheld. Installation and persistence across a reboot need a device check.

Project reference: nps/README.md. TI status and transfer references: [T5, T6].

## The entry screen and its menu

*Where commands go and how results are kept*

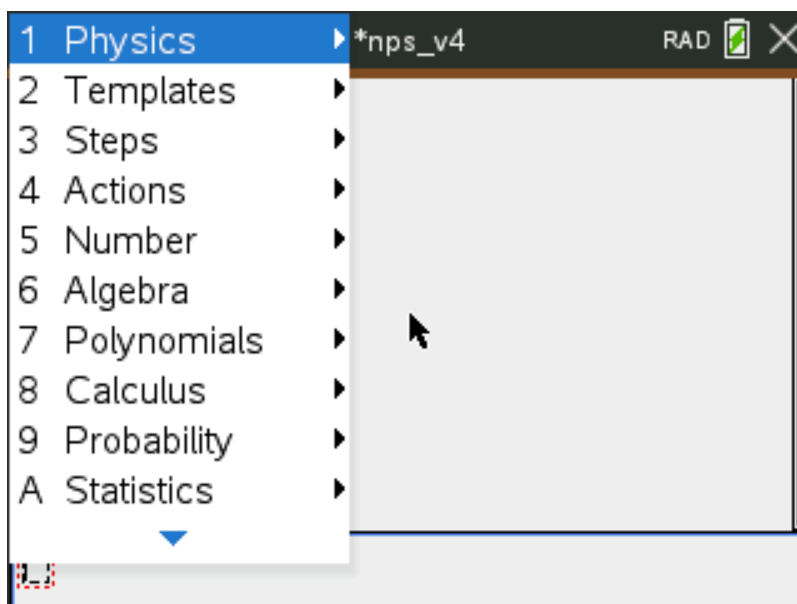


Figure 4. Retained emulator screenshot of the StepCAS palette. The down marker means more categories are below the visible list.

The bottom box is the editable entry line. Earlier inputs and outputs occupy the history above it. The title `nps_v4` identifies the document. The top system bar can show page numbers, angle mode and battery status.

1. Return to the entry screen with Escape before inserting a new command from a menu.
2. Press Menu. Use Up and Down to select a category and Enter to open it. Use Escape to back out.
3. Choose the command by its name. Many entries insert only the beginning of a call, such as `diff(`. Complete the expression, variable and closing parenthesis.
4. Review the input box before pressing Enter. A menu label describes the arguments but is not always a complete runnable example.

### The menu you see matters

TI's built-in Calculator menu is different from the StepCAS menu. StepCAS includes Physics, Templates and Steps. The separate native KhiCAS shell can have another menu layout.

Templates insert a starting structure for a fraction, power, root, derivative, integral or equation. A template helps you type. It does not guarantee that a native walkthrough exists for the resulting problem.

## 08 / LEARN &amp; USE

# Typing mathematics correctly

## *Use grouping and explicit arguments*

| Meaning              | Type                  | Watch for                                         |
|----------------------|-----------------------|---------------------------------------------------|
| Multiply             | $2*x$                 | Use * when you want to avoid ambiguous adjacency. |
| Divide a sum         | $(x+1)/(x-2)$         | Group the entire numerator and denominator.       |
| Power                | $x^3$                 | Use Right to leave the superscript slot.          |
| Square a negative    | $(-3)^2$              | This is 9. The expression $-(3^2)$ is -9.         |
| Function             | $\sin(x)$             | A function has parentheses around its argument.   |
| Several arguments    | $\text{diff}(x^3, x)$ | The comma separates the expression and variable.  |
| Equation             | $2*x+3=11$            | = states equality. It does not store a value.     |
| List                 | $[2,4,6]$             | Square brackets contain the list elements.        |
| Matrix               | $[[1,2],[3,4]]$       | Each inner list is one row.                       |
| Range in a Giac plot | $x=-5..5$             | Two dots separate the lower and upper bounds.     |

The examples show the intended expression, not a promise that each printed character corresponds to a separate keypress. The native editor turns powers and fractions into two-dimensional slots. Look at the cursor before typing a comma, plus sign or closing parenthesis.

### A useful input check

Before Enter, count the arguments and check the grouping. For  $\text{diff}((x+1)/(x-2), x)$ , the first comma comes after the whole fraction. If it is still in an exponent or denominator slot, use the arrow keys to leave that slot.

Use the negative-number key for a leading negative if the editor distinguishes it from subtraction. Use plain x for the variable rather than the multiplication symbol. Letter case can matter for stored names.

## 09 / LEARN &amp; USE

# History, copying and clearing

## *Reuse work without accidentally deleting it*

History contains inputs and displayed outputs. Use Tab and the arrow keys to move focus from the entry line into history. The focused editor or highlighted entry tells you which item will receive an action.

1. Select the old input or output you want to reuse.
2. Press Enter to copy it into the entry line. This gives you an opportunity to edit before evaluating again.
3. Change the numbers or expression, then press Enter from the entry line to calculate.
4. For more control on a focused history item, open its context menu and choose Copy to Entry Line or Delete Entry.

| Action                          | Effect                                                                                       |
|---------------------------------|----------------------------------------------------------------------------------------------|
| Del inside the entry field      | Removes the character before the cursor.                                                     |
| Ctrl+Del inside the entry field | Clears the current input.                                                                    |
| Delete a history entry          | Removes the selected input/output pair.                                                      |
| Actions > Clear History         | Asks for confirmation before clearing all history. The current unentered input is preserved. |
| Actions > Restart CAS           | Resets the algebra session. This is separate from clearing visible history.                  |

### Focus first

Ctrl+Del is not harmless everywhere. In history it is an entry-deletion action. Check that the cursor is in the bottom input field before using it to clear a line.

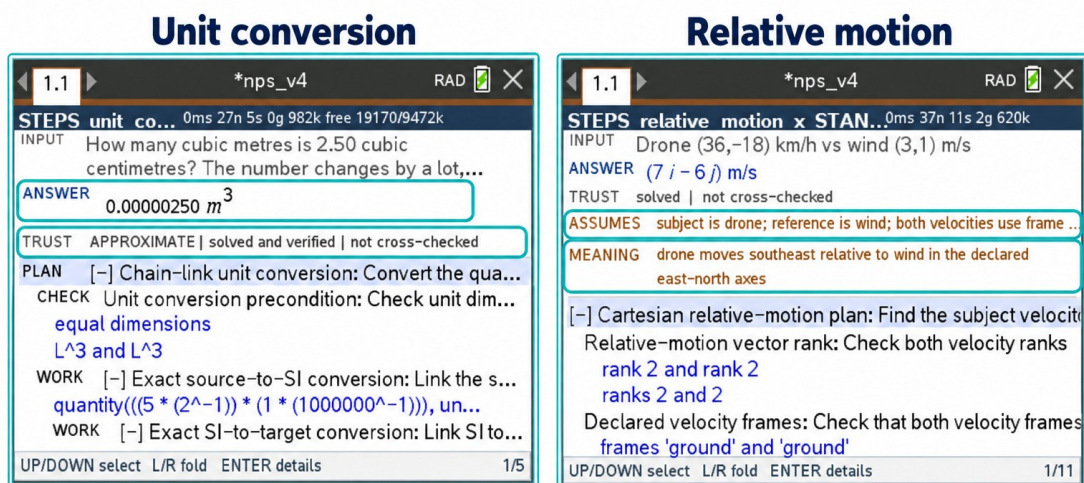
History is not an archive of all derivation trees. The last retained walkthrough can be reopened with !! during the session. Saving and reopening the document preserves history text, but a walkthrough must be computed again.

## 10 / LEARN &amp; USE

# Read a walkthrough

*Follow the answer, the evidence and the individual steps*

## Read the answer and its conditions



**ANSWER:** the value. **TRUST:** the check status. **ASSUMES:** required conditions. **MEANING:** the interpretation.

Figure 5. Annotated retained captures. These show older heading wording. Current builds can show more explicit result classifications and detail labels.

| Part of the screen | What to read                                                              |
|--------------------|---------------------------------------------------------------------------|
| Input              | The problem that was submitted. A shortened preview may omit its end.     |
| Answer             | The result, if one is available and hints permit it to be shown.          |
| Trust / status     | Whether a check ran, whether it agreed and whether work is complete.      |
| Assumptions        | Conditions under which the mathematics or physical model applies.         |
| Meaning            | The physical interpretation, when the solver supplies one.                |
| Step list          | The plan, transformations, checks and cases that make up the explanation. |

Select a step and press Enter. Current transformation details use Do for the operation, Write for the resulting expression and Why for its justification. They also retain the starting expression and conditions. Plans and checks show different information because they are different kinds of records.

### Read the status before copying

An answer supplied by Giac can coexist with some native steps. A visible value does not mean StepCAS produced a complete verified explanation of that value.

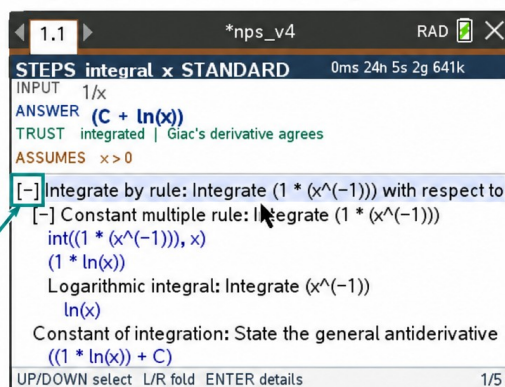
## 11 / LEARN &amp; USE

# Move through the steps

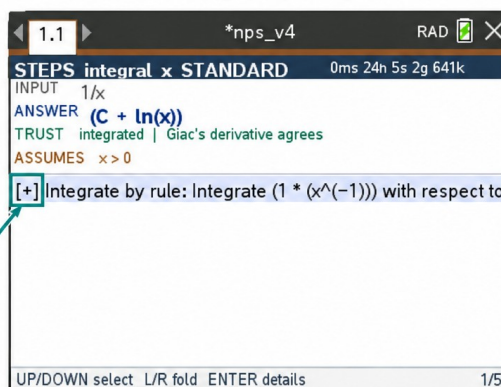
Select, expand and open the explanation you need

## Open and close a group of steps

**Expanded: the smaller steps are visible**



**Collapsed: the group stays in the list**



**Use Left and Right in the step list to fold or expand a group.**

Figure 6. Annotated older emulator captures. The integral shown assumes  $x > 0$ . The plus and minus markers identify a folded or expanded group.

| Where you are               | Controls                                                                                                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Step list                   | Up/Down selects steps. Left folds a group. Right expands it. On a row without children, Left/Right moves between steps. |
| One step open               | Up/Down scrolls its explanation. Left/Right moves to another visible step.                                              |
| Open details or result view | Enter or Escape returns to the list.                                                                                    |
| Step list                   | Escape returns to the entry screen.                                                                                     |
| List with pointer           | Click once to select a row. Click the selected row again to open it.                                                    |

A group contains smaller steps. Folding it reduces clutter without deleting any mathematics. This is especially helpful for the product rule, quotient rule or a physics solution that contains an algebra calculation.

### Nothing is recalculated when browsing

Opening a detail page, folding a branch or changing the explanation level reads the existing result. If you want to solve a changed problem, return to the entry screen and submit the new input.

## 12 / LEARN &amp; USE

# Beginner detail and hint mode

## *Control how much explanation you see*

Beginner detail and hint progression are separate settings. Beginner detail makes explanations fuller. Hint progression controls when the next part of the work and the final answer become visible.

1. For fuller explanations, open a walkthrough and press Shift+Tab. Check the header for BEGINNER. Press it again to return to STANDARD.
2. For hints, return to the entry screen and enter !h on. You can also choose Steps > Hint walkthrough (Tab next).
3. Submit a supported problem, such as !s  $2x+3=11$ . Try the next operation on paper before pressing Tab.
4. Press Tab to reveal more work. Repeat until the walkthrough is fully exposed.
5. To return to full walkthroughs for subsequent requests, enter !h off or choose Steps > Full walkthrough (all steps).

```
!h on
!s 2*x+3=11
```

### Hints are not an answer checker

You do not enter your proposed next line into the walkthrough. It reveals recorded work. A parent step's completed expression can stay hidden until its child work has been revealed.

### Commands without native steps behave differently

A Giac answer with no walkthrough can appear immediately. Hint mode cannot create missing steps or turn a limits command into a limit lesson.

If TAB result is displayed after the answer becomes available, Tab may open the complete result instead of advancing hints. Follow the footer and return to the list with Escape.

## 13 / LEARN &amp; USE

# Read long expressions and explanations

*Use the complete text when a preview is too small*

A calculator screen has limited space. A shortened input, crowded formula or trailing dots mean you may be looking at a preview. Do not copy an incomplete denominator, unit or condition from it.

| Your location                         | Open the full content                                                        |
|---------------------------------------|------------------------------------------------------------------------------|
| Walkthrough or guided physics browser | Press T, or use Actions > Read Full Text.                                    |
| Entry screen or focused history       | Use Actions > Read Full Text. The Help action also opens the reader.         |
| Answer preview says TAB result        | Press Tab to open the complete answer view when hint progression permits it. |

1. Open the full-text reader. Use Up and Down to scroll one line at a time.
2. Use Tab to move forward by a larger amount. Shift+Tab moves back.
3. Press Escape or Enter to close the reader and return to the originating view.

Full Text favors readable, wrapped text over two-dimensional notation. It retains the whole expression. The answer view and step details can still use native math boxes when the expression fits.

## Help has a special meaning inside a walkthrough

There, Help changes Standard/Beginner detail. Use T when you specifically want Full Text. The reader respects hint withholding and does not reveal future steps.

Actions > Increase Font Size and Decrease Font Size adjust the shell/history font. The walkthrough has its own fitting rules. If increasing the font makes a history expression too large, use the full reader rather than assuming the expression was lost.

## 14 / LEARN &amp; USE

# Automatic commands and explicit modes

## *Know what the next press of Enter will do*

A fresh document starts with automatic walkthrough recognition. Commands such as `solve(...)`, `diff(...)` and `int(...)` are checked for a native explanation. Other ordinary expressions go to Giac. Saved preferences can change this starting behavior.

| Input                                              | Meaning                                                            |
|----------------------------------------------------|--------------------------------------------------------------------|
| <code>!d x^3</code>                                | Differentiate this expression once using the selected variable.    |
| <code>!i x^2</code>                                | Integrate this expression once.                                    |
| <code>!s 2*x+3=11</code>                           | Solve this equation once.                                          |
| <code>!k find v, v0=5 m/s, a=3 m/s^2, t=4 s</code> | Solve this structured kinematics problem once.                     |
| <code>!d</code>                                    | Select differentiation as the persistent mode for following input. |
| <code>!i / !s / !k</code>                          | Select integration, solving or kinematics in the same way.         |
| <code>!v t</code>                                  | Use t as the selected variable for prefix commands.                |
| <code>!g</code>                                    | Clear the selected solver mode and switch to plain Giac.           |
| <code>!!</code>                                    | Reopen the most recent retained walkthrough.                       |

### Return to ordinary automatic operation

Enter `!g` on its own line. Then enter `!h off` on another line. The first clears a persistent solver mode. The second re-enables automatic recognition with full walkthroughs.

```
!g
!h off
!v x
```

The commands above are separate submissions. `!g` is a mode switch, not a prefix that evaluates text after it. In a persistent `!d` mode, type `x^3` rather than `diff(x^3,x)`, because the whole following input is treated as the expression to differentiate.

## 15 / LEARN &amp; USE

# Arithmetic, fractions and decimal answers

*Understand the number before changing its appearance*

| Try in ordinary mode        | Mathematical target                                       |
|-----------------------------|-----------------------------------------------------------|
| $2+3*4$                     | 14, because multiplication is performed before addition.  |
| $(2+3)*4$                   | 20, because parentheses change the grouping.              |
| $1/3+1/6$                   | $1/2$ , an exact fraction.                                |
| <code>evalf(1/3)</code>     | A decimal approximation to one third.                     |
| <code>sqrt(2)</code>        | An exact radical in Giac when exact evaluation is active. |
| <code>evalf(sqrt(2))</code> | Approximately 1.414213562.                                |

An exact answer represents the value without rounding. A decimal is often more convenient for measurement or plotting, but the final digits can be rounded. A fraction and its decimal approximation are not different solutions.

Use Number > Convert to Decimal or type `evalf(...)`. Complete the argument and closing parenthesis. For teaching steps, start with integer values and exact fractions so you stay inside the widest supported walkthrough coverage.

## There are separate numeric settings

TI document settings, Giac settings and the native StepCAS numeric mode are not one shared switch. The current Ki V4 walkthrough calls use the native default Exact mode. A TI status-bar setting does not establish that a native request used Decimal mode.

Physics precision is another distinction. A supplied measured value can carry significant figures even while the internal arithmetic uses exact fractions. Keep the units and the original decimal places when entering a measurement.

## 16 / LEARN &amp; USE

# Solve equations

*Find the unknown and check it in the original equation*

1. Return to ordinary automatic operation, or use the explicit !s prefix.
2. Enter `solve(2*x+3=11,x)`. Read the answer and open the steps.
3. Check that the explanation subtracts 3 and divides by 2.
4. Substitute the answer into the original equation yourself:  $2 \cdot 4 + 3 = 11$ .

```
solve(2*x+3=11, x)
!s 3*x-6=0
```

| Case      | Expected meaning                                           |
|-----------|------------------------------------------------------------|
| $2x+3=11$ | One solution: $x = 4$ .                                    |
| $x+1=x+2$ | No solution. Cancelling $x$ leaves an impossible equality. |
| $2x=2x$   | Every value is a solution within the stated domain.        |
| $x^2=9$   | The pure-square family can return $x = -3$ and $x = 3$ .   |

## Current native coverage

Linear equations in one unknown are supported. The separate quadratic family handles pure squares with exact rational roots, such as  $x^2=9$ . It does not establish a general quadratic-formula walkthrough for  $x^2+x-1=0$  or a radical-root walkthrough for  $x^2=2$ .

Giac can solve more forms than the native teaching engine. When the app supplies a CAS answer, read that label. For an equation outside the teaching coverage, you can deliberately switch to !g and run `solve(...)`, but the result is then ordinary CAS output.

## 17 / LEARN &amp; USE

# Rearrange, simplify, expand and factor

*Choose the operation that matches your goal*

| Goal                       | Example                          | Expected form                |
|----------------------------|----------------------------------|------------------------------|
| Isolate a formula variable | <code>rearrange(V=I*R,I)</code>  | $I = V/R$ , with R not zero. |
| Combine like terms         | <code>simplify(2*x+3*x)</code>   | $5*x$ .                      |
| Multiply out brackets      | <code>expand((x+1)*(x+2))</code> | $x^2+3*x+2$ .                |
| Write a product            | <code>factor(x^2+5*x+6)</code>   | $(x+2)*(x+3)$ .              |

Simplifying changes the form of an expression. Solving finds values that make an equation true. Factoring is useful for seeing products and zeros, but `factor(...)` by itself does not ask for an equation's solution set.

1. Choose Algebra > Simplify, Expand or Factor. Complete the call and press Enter.
2. Open a transformation and read Do, Write and Why.
3. For a factorization, multiply the result back out to check that it matches the original expression.

## Where the native explanation stops

Formula rearrangement supports a single occurrence of the target symbol through supported inverse operations. Polynomial rewrite steps cover exact arithmetic, common factors and supported monic quadratic factoring. General rational simplification, decimal inputs, radicals and higher-degree factoring can exceed that coverage.

An expression already in the requested form can finish without a transformation. This is a valid outcome. Extra steps should not be invented just to make the list longer. Read any conditions introduced by division or reciprocals.

## 18 / LEARN &amp; USE

# Differentiate a function

*Find a rate of change with named rules*

```
diff(x^3,x)
!d sin(x)
diff((x+1)/(x-2),x)
```

1. Use Calculus > Derivative or type `diff(x^3,x)`. The first argument is the function. The second is the variable.
2. For  $x^3$ , expect  $3x^2$ . Open the power-rule step to see how the exponent is used.
3. Try a product or quotient and expand its group. The smaller steps calculate the derivatives needed by the main rule.
4. Read any domain restrictions. A denominator must remain nonzero where the formula is used.

| Function      | Mathematical derivative               |
|---------------|---------------------------------------|
| $x^3$         | $3x^2$                                |
| $\sin(x)$     | $\cos(x)$ , using radians             |
| $\exp(2x)$    | $2\exp(2x)$                           |
| $(x+1)/(x-2)$ | $-3/(x-2)^2$ , for $x$ not equal to 2 |

## Which rules have native steps

The implementation covers constants, the variable, sums, constant multiples, products, quotients, integer powers, the chain rule and supported  $\sin$ ,  $\cos$ ,  $\tan$ ,  $\exp$ ,  $\ln$  and  $\sqrt{\phantom{x}}$  forms. Particular domains and input shapes can still cause a refusal.

Implicit differentiation, partial-derivative lessons and higher derivatives in a single request are outside the native walkthrough family. A different letter works when explicitly supplied, for example `diff(t^3,t)`. For prefix entry, select it with `!v t` and restore `!v x` when finished.

## About the input appearance

The current input may show `diff(...)` as a function call. A stacked editable  $d/dx$  template is not established as working in this build. The textbook-style input discussed earlier has not been implemented by this manual.

## 19 / LEARN &amp; USE

# Integrate a function

*Find an antiderivative and remember the constant*

```
int(x^2,x)
int(sin(x),x)
int(exp(2*x),x)
```

| Integrand  | Mathematical antiderivative               |
|------------|-------------------------------------------|
| $x^2$      | $x^3/3 + C$                               |
| $\sin(x)$  | $-\cos(x) + C$                            |
| $\exp(2x)$ | $\exp(2x)/2 + C$                          |
| $1/x$      | $\ln(x) + C$ on the stated $x > 0$ branch |

An indefinite integral describes a family of functions.  $C$  is the constant of integration. Differentiating  $C$  gives zero, so every value of that constant has the same derivative.

1. Choose Calculus > Integral and complete `int(x^2,x)`.
2. Read the antiderivative and open the rule steps. The native integrator checks its answer by differentiating it.
3. For a linear argument such as  $\exp(2x)$ , look for the substitution or scale correction.
4. Read the assumptions for reciprocal and logarithmic cases before using the answer.

## Do not silently discard a domain condition

The positive branch  $\ln(x)$  is not a claim about negative  $x$ . The general real antiderivative of  $1/x$  on intervals avoiding zero is  $\ln(\text{abs}(x)) + C$ . Use the branch and conditions actually reported by StepCAS.

## Current native limits

The supported lessons include sums, constant multiples, integer powers, reciprocals and  $\sin$ ,  $\cos$  or  $\exp$  with supported linear arguments. Integration by parts, nonlinear substitution and general integrals involving  $\tan$ ,  $\ln$ ,  $\sqrt{x}$  or rational exponents are outside this family.

For a definite integral such as `int(x^2,x,0,1)`, the expected value is  $1/3$ . Definite-integral walkthroughs are not supported. Use ordinary Giac deliberately when you need the calculation and read the result as CAS output.

## 20 / LEARN &amp; USE

# Limits, sums and products

*Use Giac for the value and recognize the absence of native lessons*

1. Enter !g on its own line to select ordinary Giac.
2. Enter a complete limit, sum or product expression from the examples below.
3. Check the approach value, index and bounds carefully.
4. When finished, enter !h off to restore automatic full walkthrough recognition.

| Type                                  | Meaning                                    | Mathematical target |
|---------------------------------------|--------------------------------------------|---------------------|
| <code>limit(sin(x)/x,x,0)</code>      | Approach zero from both sides, in radians. | 1                   |
| <code>limit((x^2-1)/(x-1),x,1)</code> | Approach the removable hole at $x=1$ .     | 2                   |
| <code>limit(1/x,x,0,1)</code>         | Approach zero from the right.              | Positive infinity   |
| <code>limit(1/x,x,0,-1)</code>        | Approach zero from the left.               | Negative infinity   |
| <code>sum(2^(-n),n,1,10)</code>       | Add terms $n=1$ through $n=10$ .           | 1023/1024           |
| <code>product(n,n,1,4)</code>         | Multiply 1, 2, 3 and 4.                    | 24                  |

A limit asks what values are approached. It does not necessarily ask for the value at the point itself. In the second example, the original fraction is undefined at  $x=1$  even though the limit exists.

## Native limit walkthroughs are not implemented

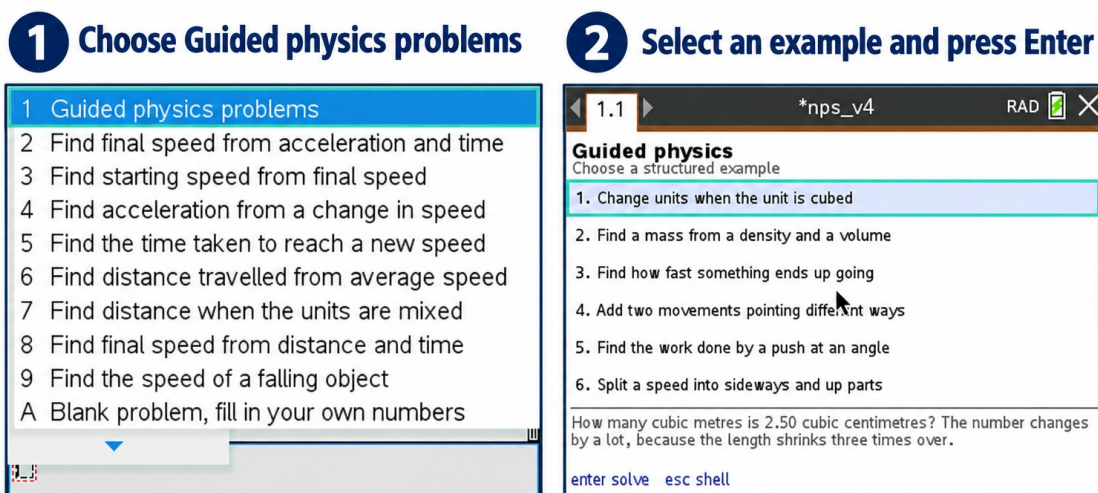
The Calculus menu exposes `limit(...)` through Giac. It does not provide a limit-rule derivation, a continuity lesson or L'Hopital steps. The sum and product examples here are also ordinary CAS calculations.

Use the word infinity for an infinite bound in Giac syntax. An unevaluated expression is not a proof that a limit or sum does not exist. Check the command, domain and backend response. Source for Giac limit direction and sequence syntax: [G1, G2].

# Guided physics

*Start with a supplied problem before entering your own model*

## Opening a guided physics example



**The browser runs supplied examples. Read the problem description before solving.**

Figure 7. Annotated retained emulator captures of the Physics menu and guided browser. Current labels are matched against the source.

1. Return to the entry screen. Choose Menu > Physics > Guided physics problems.
2. Use Up and Down to select an example. Read the description at the bottom. Press T if it is shortened.
3. Press Enter to run the selected example. It opens the usual walkthrough viewer.
4. Use Escape to return to the entry screen. Reopen the Physics browser from its menu when you want another example.

### These are supplied examples

The browser is not a general editable physics worksheet. Its stored examples cover unit conversion, density, kinematics, vectors, work, components, an incompatible-unit refusal, catch-up and relative motion. The selected example runs with its prepared quantities.

For your own one-dimensional kinematics numbers, use the Physics menu templates or !k input described next. Other structured solver capabilities exist in the native interface but do not all have general input forms in the current document.

## 22 / LEARN &amp; USE

# Enter your own motion problem

*Use constant acceleration along one axis*

```
!k find v, v0 = 5 m/s, a = 3 m/s^2, t = 4 s
```

| Symbol | Meaning                                       | Typical unit     |
|--------|-----------------------------------------------|------------------|
| $v_0$  | Initial velocity at the start of the interval | m/s              |
| $v$    | Final velocity                                | m/s              |
| $a$    | Constant acceleration                         | m/s <sup>2</sup> |
| $t$    | Elapsed time                                  | s                |
| $x$    | Displacement along the chosen axis            | m                |

1. Decide which direction is positive. Write the unknown after find.
2. Enter the known quantities with units. Separate quantities with commas or the supported semicolon syntax.
3. Press Enter. Read the assumptions and the chosen equation.
4. For the example, the model uses  $v = v_0 + a \cdot t$ . Substitution gives  $5 + 3 \cdot 4 = 17$  m/s.

## The menu says speed and distance in places

The signed one-axis model uses velocity and displacement. A negative result can mean motion in the negative direction. If an object reverses direction, total distance traveled and signed displacement are different quantities.

Try changing only  $v_0$  to 18 km/h. That is 5 m/s, so the same acceleration and time still give 17 m/s. The conversion should be part of the explanation.

The planner can use an intermediate quantity when one equation is not enough. It still has a limited collection of constant-acceleration equations and algebra routes. Missing information, an unsupported square-root step or an incompatible unit can stop it. The !k entry is not a solver for arbitrary word problems, projectile motion or variable acceleration.

# Units, volume and density

*Keep the unit attached to the number*

| Guided example                           | What the mathematics says                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Change units when the unit is cubed      | $2.50 \text{ cm}^3 = 0.00000250 \text{ m}^3$ . Each centimetre contributes a factor of 1/100, so volume has a factor of 1/1000000. |
| Find a mass from a density and a volume  | $2 \text{ g/cm}^3$ multiplied by $3 \text{ cm}^3$ gives 6 g, or 0.006 kg.                                                          |
| See what happens when units cannot match | 3 m cannot be converted to seconds without a physical relationship. A refusal is the correct result.                               |

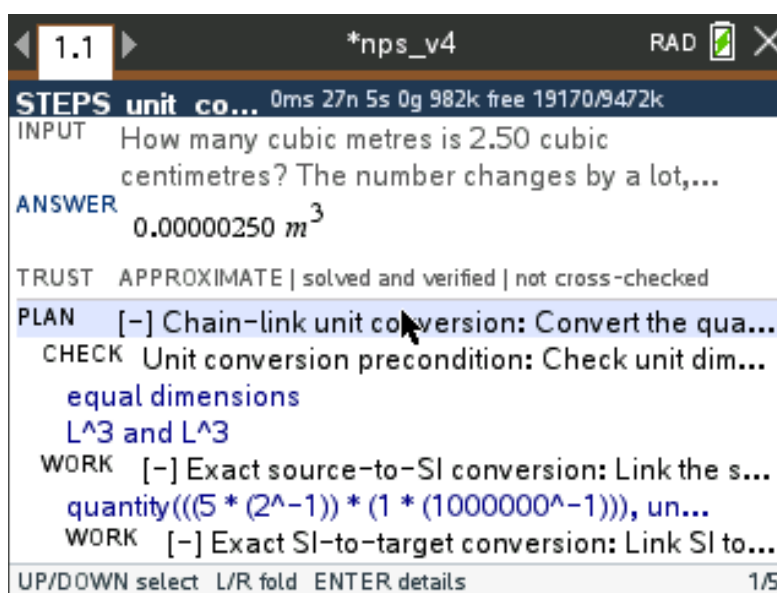


Figure 8. Retained emulator capture of the volume conversion. Current builds may present clearer Do, Write and Why labels inside details.

Density means mass per volume:  $\text{mass} = \text{density} * \text{volume}$ . The units explain why cubic centimetres cancel in the example. A bare number such as 3 does not communicate whether you mean cubic centimetres, metres or kilograms.

## Exact scale is different from measurement precision

The centimetre-to-metre scale is exact. The written value 2.50 carries measured precision. StepCAS can preserve that distinction and round the displayed measurement at the end. Trailing zeros can carry information.

The native conversion family uses compatible dimensions and exact multiplicative scales. Temperature conversions requiring an offset are outside that family. The current browser runs the supplied density and conversion examples rather than exposing a general editable form.

## 24 / LEARN &amp; USE

# Vectors, components and work

## *Match directions, units and reference frames*

A vector has components. In two dimensions,  $i$  names the horizontal basis direction and  $j$  the vertical one. A reference frame tells you what those directions mean. Adding components only makes sense when their units and frames agree.

| Guided example                            | What to look for                                                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Add two movements pointing different ways | (0.00120, 0.0020) km becomes (1.20, 2.0) m. Add (5,5) m component by component. The mathematical sum is (6.20,7.0) m before final precision presentation.        |
| Find the work done by a push at an angle  | Force (2,-3) N dotted with displacement (-1,4) m gives $2*(-1)+(-3)*4 = -14$ J.                                                                                  |
| Split a speed into sideways and up parts  | At 10 m/s and 30 degrees above the horizontal, $x = 10*\cos(30 \text{ degrees})$ and $y = 10*\sin(30 \text{ degrees})$ . The targets are $5*\sqrt{3}$ and 5 m/s. |

Negative work means the force component along the displacement opposes the motion. It is not a negative distance and it is not automatically an error.

1. Select a guided example and read its problem description.
2. Open the unit-conversion or component step first if the numbers seem surprising.
3. Check the stated frame and the angle unit.
4. Read the sign interpretation before copying the final value.

### Where the current models stop

These are selected Cartesian and constant-force cases. They do not establish arbitrary rotating frames, position-dependent force work, line integrals or general three-dimensional vector lessons. A pretty vector display is not evidence that all of those operations are implemented.

## 25 / LEARN &amp; USE

# Catch-up and relative motion

## *Distinguish elapsed time from clock time*

In the catch-up example, Atlas starts at position zero at time zero and moves at 2 m/s. Boreal starts at the same position 5 seconds later and moves at 4 m/s. Both motions use constant velocity.

```
Atlas:    position = 2*t
Boreal:   position = 4*(t-5), while t >= 5
Meeting:  2*t = 4*(t-5)
```

Solving gives  $t = 10$  s measured from Atlas's start. Boreal has run for 5 s. Both are at 20 m. Checking the active time interval matters. A meeting calculated before Boreal starts would not satisfy the prepared model.

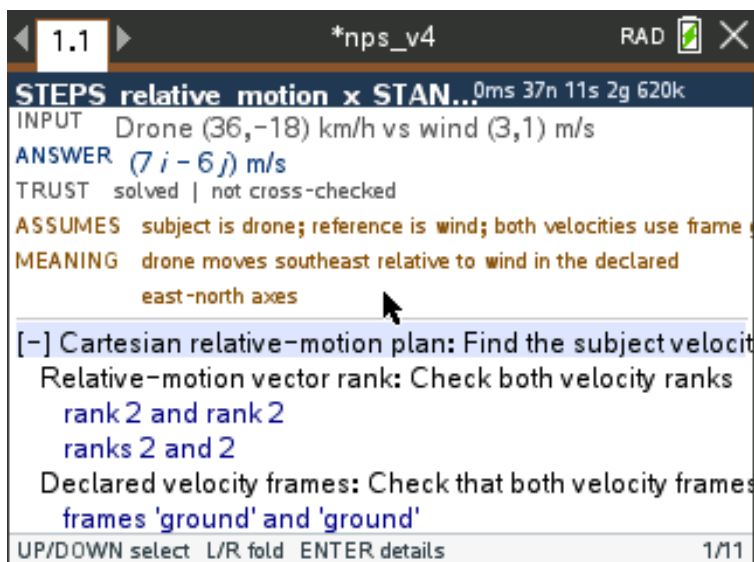


Figure 9. Retained relative-motion capture. Its answer is a velocity relative to the moving reference, not a speed magnitude.

For relative motion, the drone's ground velocity (36,-18) km/h is (10,-5) m/s. Subtract the wind velocity (3,1) m/s to obtain (7,-6) m/s relative to the wind. Positive horizontal and negative vertical components explain the southeast interpretation in the declared east-north axes.

### Current access and scope

Both examples are available through the guided browser. They use compatible declared frames. They do not infer a coordinate transformation, solve arbitrary accelerating catch-up problems or accept a paragraph of prose as input.

## 26 / LEARN &amp; USE

# Matrices and row operations

*Read rows as lists and follow every change*

```
rref([[1,2,5],[3,4,11]])
ref([[1,2],[3,4]])
det([[1,2],[3,4]])
```

A matrix is a list of row lists. In the first example, the final column holds the right-hand sides of  $x+2y=5$  and  $3x+4y=11$ . Reduced row echelon form should be  $[[1,0,1],[0,1,2]]$ , so  $x=1$  and  $y=2$ .

| Command              | Meaning                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| <code>ref(M)</code>  | Row echelon form. Pivots move right as you go down. Entries below each pivot are zero. Pivots need not be 1. |
| <code>rref(M)</code> | Reduced row echelon form. Pivots are 1 and are the only nonzero entries in their columns.                    |
| <code>det(M)</code>  | The determinant of a square matrix. The example above has determinant -2.                                    |

Native matrix walkthroughs record actual backend row operations and check their effect. A determinant explanation must also track how swaps and scaling affect the determinant. Row addition has a different effect from multiplying a row.

## Current teaching envelope

`ref` and `rref` accept exact rational matrices with up to 4 rows and 6 columns. `det` accepts square matrices of orders 1 through 4. Decimal, symbolic and complex entries are outside these walkthrough families. Intermediate arithmetic can also exceed checked bounds.

## Installed-build qualification

The current source includes these native command handlers, but the family catalog still marks physical keypad qualification pending. If your package lacks the required matrix backend support, record the refusal. Do not infer a successful trace from a menu entry alone.

The Matrix & Vector menu also exposes inverse, eigenvalues, decompositions and other Giac operations. Their presence does not imply native step coverage.

## 27 / LEARN &amp; USE

# Integers and counting

*Use exact whole-number inputs for these walkthroughs*

| Command                 | Example                      | Mathematical target                     |
|-------------------------|------------------------------|-----------------------------------------|
| Integer quotient        | <code>iquo(17,5)</code>      | 3                                       |
| Integer remainder       | <code>irem(17,5)</code>      | 2                                       |
| Greatest common divisor | <code>gcd(84,30)</code>      | 6                                       |
| Factorial               | <code>factorial(5)</code>    | 120                                     |
| Ordered selections      | <code>perm(5,2)</code>       | 20                                      |
| Unordered selections    | <code>comb(5,2)</code>       | 10                                      |
| Prime test              | <code>is_prime(29)</code>    | 2 means proven prime in this interface. |
| Next prime              | <code>nextprime(29)</code>   | 31                                      |
| Prime factors           | <code>ifactor(84)</code>     | $2^2 \cdot 3 \cdot 7$                   |
| Modular power           | <code>powmod(2,10,17)</code> | 4                                       |

A factorial multiplies the positive integers up to  $n$ . A permutation counts ordered selections. A combination counts selections where order does not matter. The distinction changes the answer even when the two input numbers are the same.

Native integer methods expect literal integer arguments. Use `gcd(84,30)`, not `gcd(7*12,30)`, if you want to stay inside the declared literal-input family. A mathematically equivalent compound argument can be outside that family.

## Bounds are deliberate

The native family allows factorial through 100 and permutations/combinations with  $0 \leq k \leq n \leq 100$ . Trial-division prime and factorization lessons are bounded around one million. Division and gcd inputs are bounded around one billion. Negative inputs are supported for gcd but generally refused by the other listed methods. Resource limits still apply.

The prime-test result is a status code. Do not read 2 as “two factors.” Other probability and number commands can remain ordinary Giac operations. Device qualification for the newer integer walkthroughs remains pending in the catalog.

## Ordinary Giac calculations

### *Use the broader algebra engine deliberately*

Enter !g to switch to ordinary Giac. In this mode, StepCAS does not request a native walkthrough for every recognized command. The engine can handle many more algebra, list, statistics and plotting operations than the teaching core.

| Example          | Purpose or mathematical target                                     |
|------------------|--------------------------------------------------------------------|
| a:=5             | Store 5 under the name a in the Giac session.                      |
| a+2              | Use that stored value. The target is 7.                            |
| purge(a)         | Remove the assignment when you want a to be symbolic again.        |
| subst(x^2+1,x,3) | Substitute 3 for x. The target is 10.                              |
| sort([5,1,3])    | Sort a list into ascending order.                                  |
| mean([2,4,6])    | Arithmetic mean. The target is 4.                                  |
| median([2,4,9])  | Middle value. The target is 4.                                     |
| conj(2+3*i)      | Complex conjugate, normally $2-3i$ when $i$ is the imaginary unit. |

#### Stored variables can surprise you

If  $x$  was assigned a number, a later symbolic expression can evaluate using that number. Check assignments before blaming the algebra. Use `purge(x)` when appropriate. Restart CAS is broader and can discard session definitions.

The Statistics menu includes list manipulation, plots and regression. The Probability menu includes distributions and tests. A probability density, a cumulative probability and an inverse cumulative probability answer different questions. Look up the exact signature before using one.

TI's document variables and Giac's stored values are separate contexts. Saving the document's visible history is not a guarantee that every Giac assignment has been exported. Use the save guidance later in this manual. Source: Giac reference [G0, G3].

## 29 / LEARN &amp; USE

# Angles and trigonometry

*Check the interpretation before trusting a sine or cosine*

Degrees describe a full turn as 360. Radians describe it as  $2\pi$ . The standard calculus identities  $d(\sin(x))/dx = \cos(x)$  and the limit  $\sin(x)/x = 1$  at zero assume radians.

| Purpose                                  | Expression to use                               |
|------------------------------------------|-------------------------------------------------|
| A 30-degree angle in a radian expression | $\sin(\pi/6)$                                   |
| Convert 30 degrees to radians            | $30\pi/180$                                     |
| Convert $\pi/3$ radians to degrees       | $(\pi/3)*180/\pi$                               |
| Check a familiar value                   | In radian operation, $\sin(\pi/2)$ should be 1. |

The RAD or DEG indicator in the system bar describes a TI setting. StepCAS also uses Giac and native mathematical routines. Do not assume every layer follows that indicator. Use a known-value check in the mode where you are actually calculating.

## For the manual's calculus examples

Use radian conventions. For the guided 30-degree vector example, the request explicitly supplies degrees. That is why its component calculation uses a 30-degree angle rather than 30 radians.

TI document settings are available through Home > Settings > Document Settings. Changing them affects TI applications as described by TI. This manual does not prescribe an unverified Giac angle-setting shortcut for the installed build.

If a numerical trigonometric result seems wrong, check three things in order: the input expression, the angle convention and any stored variable assignments. Changing the number of displayed decimal places cannot fix a degree/radian mistake.

TI setting reference: [T7].

## Graphs and the separate native shell

*Recognize the screen you have entered*

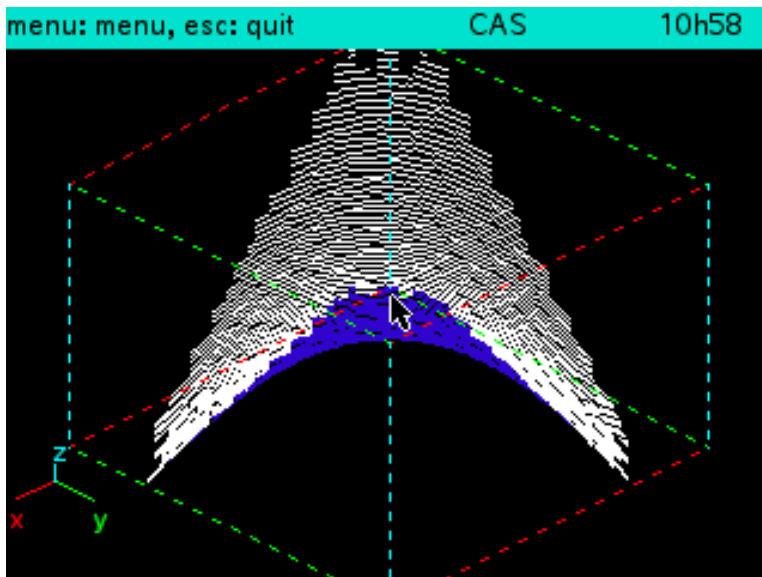


Figure 10. Retained native Giac 3D graph capture. Its own menu and Escape hints appear at the top. This is not the StepCAS step viewer.

StepCAS exposes Giac plots in its Plots menu. A plotting command can open a separate native graphics view. Use that view's on-screen controls rather than the walkthrough's fold and hint keys.

```
plot(x^2, x=-5..5)
```

1. Use ordinary Giac mode for this introductory plot. Type the command above and press Enter.
2. If the graph opens, look at the visible x and y range. A blank view can mean the curve is outside the window.
3. Open the graph's Menu for the available view controls. Use its displayed Escape behavior to return.
4. Return to the StepCAS entry line before choosing a walkthrough or guided problem.

The Plots menu also lists area, parametric, polar, slope-field, differential-equation, density, contour and implicit plotting commands. Their arguments differ. Use the Giac reference for the command you choose.

### Open Shell is an advanced action

Actions > Open Shell inserts \*. Submitting it enters the additional native KhiCAS shell. It is not required for ordinary StepCAS input. Actions > Open Script Editor is also a separate environment with its own controls.

Graphs can illustrate behavior but a picture is not a symbolic proof, and a screenshot of a 3D surface does not establish every plotting command works in the current package. Giac graph reference: [G4].

## 31 / LEARN &amp; USE

## The calculator outside StepCAS

*Use the built-in applications without confusing their results*

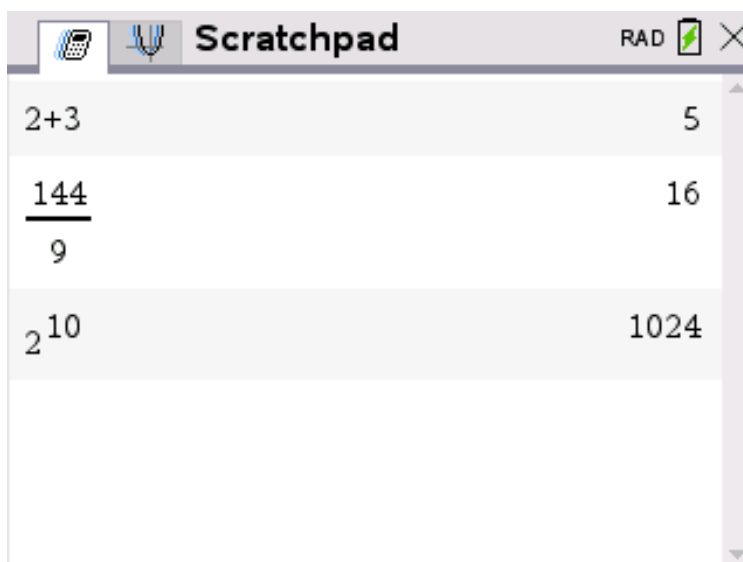


Figure 11. Retained emulator capture of TI's built-in Scratchpad Calculate. Its layout and available commands differ from StepCAS.

| TI application               | Use it for                                                                |
|------------------------------|---------------------------------------------------------------------------|
| Scratchpad Calculate / Graph | Quick calculations or graphs without first creating a worksheet document. |
| Calculator                   | Calculations inside a saved TI document.                                  |
| Graphs / Geometry            | Graphing functions and working with geometric objects.                    |
| Lists & Spreadsheet          | Organizing columns of values and formulas.                                |
| Data & Statistics            | Visualizing and analyzing data.                                           |
| Notes                        | Writing explanations and mathematical notes.                              |
| Vernier DataQuest            | Supported sensor and data-collection activities.                          |

Use Home to choose a built-in app. A TI document can contain problems and pages. Ctrl+Left/Right moves between document pages. Doc provides file and page commands. These are system document operations, not steps in a mathematical derivation.

### CX II versus CX II CAS

The non-CAS CX II's built-in Calculator has different symbolic capabilities from the CAS model. StepCAS uses its own Giac engine, so a working StepCAS limit or integral does not mean the same syntax is available in TI Scratchpad.

Sources: TI Scratchpad and document guide [T4, T8].

## Save, reopen and back up

### *Preserve your input and understand what is not saved*

1. Return to the StepCAS entry screen when convenient. Press Ctrl+S, or use Doc > File > Save.
2. Complete the system save dialog if one appears. Use a clear name for an ordinary study document.
3. Before modifying the application package or moving it to another folder, preserve the installed copy and its matching sidecars.
4. Use Browse or Recent to reopen the document. Check that your history and pending input are present.

| Item                                   | Current StepCAS document behavior                                          |
|----------------------------------------|----------------------------------------------------------------------------|
| History text                           | Saved and restored.                                                        |
| Unentered input                        | Saved and restored.                                                        |
| Selected variable and solver mode      | Saved and restored.                                                        |
| Standard/Beginner and hint preferences | Saved and restored.                                                        |
| Complete last derivation tree          | Not serialized. Recompute the problem after reopening.                     |
| All Giac variables and definitions     | Not established by the document save routine. Export separately if needed. |

The Actions menu exposes Save Variables and Read Variables commands. They are Giac file operations, separate from Ctrl+S. The Save Variables item inserts an unfinished call. Do not press Enter on that fragment or assume a successful document save exported all variables.

#### **Back up the working package as a set**

The document, native module and their matching sidecars belong together. Keep runtime files separate from ordinary personal worksheet backups. A renamed or moved application copy has not automatically passed the package integrity and loading checks.

Transfer with a compatible TI desktop tool or CX II Connect as supported by your setup. Verify the destination and open a copied ordinary document before relying on the backup. A saved file and a working runtime after reboot are separate checks. Sources: [T6, T9].

## Battery, display and routine care

### *Keep the calculator ready without disturbing the app installation*

Charge the handheld through the supported USB connection or charger. If it has been stored for a long time, allow it to charge before concluding it is broken. The battery indicator and Handheld Status provide more information than the color of a cable or the presence of a screen alone.

1. Save your document before powering off or disconnecting a transfer.
2. Use Ctrl+Home / On to power off. Use Home / On to turn it on again.
3. Use Ctrl+Plus or Ctrl+Minus for the display brightness/contrast adjustment supported by the handheld.
4. For text that is hard to read inside StepCAS, first try the Actions font controls or the Full Text reader.

#### **Operating-system updates affect this setup**

StepCAS depends on a compatible native runtime. Do not treat a generic recommendation to install the latest TI OS as an application maintenance step. Confirm runtime compatibility and retain backups before any OS change.

Do not use Reset Storage, delete the runtime folder or remove recovery files as routine troubleshooting. Those actions are different from clearing a mistyped expression. If the calculator stops responding, record the state and start with the recovery guidance on the next pages.

Do not connect or disconnect a cable while an active transfer is being written. For battery replacement, charging faults or physical damage, use the official handheld instructions for the exact model.

Sources: TI power, charging and handheld setup [T10, T11].

## 34 / LEARN &amp; USE

# When the answer surprises you

*Check meaning and input before trying a larger problem*

| Symptom                         | First check                                               | What to do                                                                     |
|---------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------|
| The value is a fraction         | Exact evaluation may be active.                           | Use <code>evalf(...)</code> for a decimal approximation.                       |
| A trig answer seems wrong       | Degrees, radians or a stored variable.                    | Test a familiar angle in the same calculator mode.                             |
| There are two answers           | A square equation may have two roots.                     | Check both in the original equation.                                           |
| No solution                     | The reduced equation may be inconsistent.                 | Read the check rather than treating this as a crash.                           |
| Every value                     | Both sides may be identical.                              | This is different from one particular solution.                                |
| CAS answer                      | The value came from the backend.                          | Do not infer a complete native walkthrough.                                    |
| Not cross-checked               | The extra backend check did not establish agreement.      | Read the available native checks and conditions.                               |
| DISAGREES / verification failed | A check did not support the result.                       | Keep the exact input and message. Do not copy the disputed result as verified. |
| Unsupported form                | The lesson family does not cover the input.               | Use a supported example or ordinary Giac intentionally.                        |
| Resource limit reached          | An arithmetic, memory or work budget stopped the request. | Reduce the problem or expression. A prefix of steps is not completion.         |

A partial derivation can contain valid earlier steps even when the whole problem did not finish. Read the final status. A green agreement message is useful evidence, but it does not make all displayed methods independent or all possible inputs supported.

## When asking for help

Record the exact expression, selected variable, persistent mode, hint setting, package identity from `!m` and the full error text. A photo of the complete screen is much more useful than "it did not work."

# When navigation or loading goes wrong

*Recover from the smallest problem first*

| Symptom                                       | Recovery                                                                                                                                     |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| I cannot type                                 | You may be in a step, reader or guided browser. Press Escape to return to the entry screen, then select the bottom field.                    |
| The menu inserts nothing                      | Close the walkthrough or reader before using an input template. The hidden entry is protected while another view is open.                    |
| The answer is hidden                          | Check for hint mode. Press Tab to reveal the remaining work.                                                                                 |
| A formula is clipped                          | Use T in a walkthrough or Actions > Read Full Text from the shell.                                                                           |
| Enter keeps differentiating                   | A bare !d selected persistent mode. Enter !g, then !h off on the next line to restore ordinary automatic operation.                          |
| !! says no steps yet                          | No retained walkthrough exists in this session. Submit the problem again.                                                                    |
| The app says incompatible or integrity failed | Stop mixing files. Restore a matching document/module/sidecar package.                                                                       |
| The title shows the wrong application         | Use Home > Current or Browse to open the intended StepCAS document.                                                                          |
| A calculation appears stuck                   | Try the current view's cancellation control and allow time for the backend to respond. Cancellation is cooperative and may not be immediate. |

If you can still reach Home, save what you can and record the problem before reopening the app. If the whole handheld is unresponsive, follow TI's model-specific recovery instructions. A forced restart can lose unsaved work and the runtime may need to be checked afterward.

## Do not make a loading problem harder to diagnose

Do not repeatedly replay file transfers after an uncertain failure. Check what arrived first. Do not erase storage or update the OS just because one StepCAS command failed.

## What currently has a native walkthrough

*Use this as a coverage guide, not a device acceptance certificate*

| Family                       | Current scope                                                                    |
|------------------------------|----------------------------------------------------------------------------------|
| Linear equations             | One unknown, supported linear grammar and exact arithmetic.                      |
| Pure-square quadratics       | No linear term, real cases and exact rational roots.                             |
| Formula rearrangement        | One occurrence of the requested symbol with supported inverses.                  |
| Polynomial rewrites          | Simplify, expand and supported factoring with exact coefficients.                |
| Derivatives                  | First derivatives by supported elementary rules.                                 |
| Indefinite integrals         | Supported elementary forms and linear-argument substitution.                     |
| Integer methods              | Bounded literal inputs for the listed division, gcd, counting and prime methods. |
| Matrices                     | Small exact rational ref, rref and determinant cases.                            |
| One-axis kinematics          | Constant acceleration with structured known quantities.                          |
| Unit conversion and density  | Compatible multiplicative units and mass-volume-density relations.               |
| Vectors and work             | Selected component, magnitude/direction and constant-force cases.                |
| Catch-up and relative motion | Declared constant-velocity / compatible-frame examples.                          |

### Available in the engine is not the same as editable in the UI

The general !k input supports your own kinematics quantities. Several other physics families are exposed through supplied browser examples or structured native calls, not a complete beginner input form.

Limits, definite-integral lessons, general quadratic methods, implicit or partial differentiation lessons and broad multiple/contour-integral walkthroughs are not established native features. Giac may calculate an answer for some of them. Newer families remain marked unreleased with physical qualification pending in the catalog.

Source of truth: [nps/catalog/families.md](#) and the current command/UI implementation.

## Practice with the controls

### *Learn one navigation habit at a time*

1. Open StepCAS, type  $!s\ 2*x+3=11$  and press Enter. Find the answer before opening any detail.
2. Select the division step. Open it with Enter. Read the operation and the resulting line.
3. Switch to Beginner detail with Shift+Tab. Identify what explanation was added.
4. Return to the entry screen. Enter  $!h\ on$ , then solve another equation with  $!s\ 3*x-6=0$ .
5. Before each Tab, write what you think comes next. Continue until the answer appears.
6. Return to the entry screen. Enter  $!h\ off$ . Reopen the retained walkthrough with  $!!$ .
7. Open Guided physics and select the incompatible-unit example. Find the reason for its refusal.
8. Save the document. Write down which things the Save routine preserves and which must be recomputed.

#### **You have the navigation basics when...**

You can find the entry field, open a step, return to the list, return to the entry screen and read a long expression without losing your place. Speed is less important than recognizing which view has focus.

If an exercise behaves differently from this description, check the installed package and persistent mode. Do not practice around a mismatch until it becomes a habit.

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## Practice problems and answer checks

*Try these on paper before looking at the final column*

| Task                    | Input                                 | Answer check                                     |
|-------------------------|---------------------------------------|--------------------------------------------------|
| Linear equation         | <code>!s 5*x+4=19</code>              | $x=3$ , because $5*3+4=19$ .                     |
| Pure square             | <code>solve(x^2=16,x)</code>          | $x=-4$ or $x=4$ .                                |
| Expand                  | <code>expand((x+3)*(x+2))</code>      | $x^2+5x+6$ .                                     |
| Factor                  | <code>factor(x^2+5*x+6)</code>        | $(x+2)*(x+3)$ .                                  |
| Differentiate           | <code>diff(4*x^3,x)</code>            | $12x^2$ .                                        |
| Integrate               | <code>int(3*x^2,x)</code>             | $x^3+C$ .                                        |
| Greatest common divisor | <code>gcd(48,18)</code>               | 6.                                               |
| Combination             | <code>comb(6,2)</code>                | 15.                                              |
| Determinant             | <code>det([[2,1],[1,3]])</code>       | 5.                                               |
| Limit in ordinary Giac  | <code>limit((x^2-4)/(x-2),x,2)</code> | 4, although the fraction is undefined at $x=2$ . |

### Explain the difference

Why can  $x^2=16$  have two roots? Why does an indefinite integral include  $C$ ? Why does an input containing centimetres cubed require three factors of the length conversion? Why does “CAS answer” tell you less about teaching steps than “solved and verified”?

### Physics check

For constant acceleration with  $v_0=2$  m/s,  $a=4$  m/s<sup>2</sup> and  $t=3$  s, the target final velocity is 14 m/s. Choose a positive direction and keep it consistent. For density 2 g/cm<sup>3</sup> and volume 3 cm<sup>3</sup>, the target mass is 6 g.

#### Expected values, not new device receipts

These answers have been checked mathematically for this manual. They are not claims that every lesson was freshly entered and rendered on the handheld during preparation.

## 39 / LEARN &amp; USE

# Keep this page beside the calculator

*A compact navigation and command card*

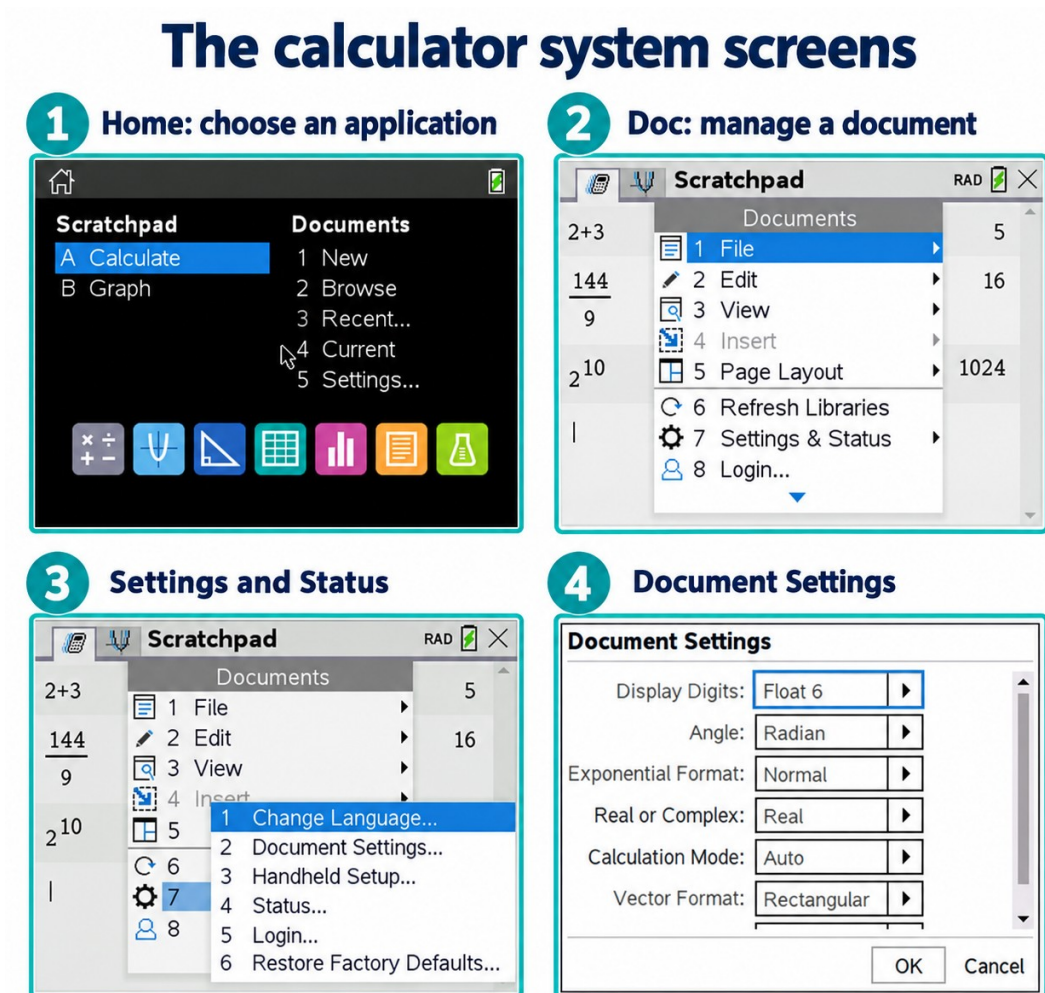
| I want to...                | Do this                                                            |
|-----------------------------|--------------------------------------------------------------------|
| Open StepCAS                | Home > Browse > installed nps_v4 document.                         |
| Calculate                   | Select the entry line, type a complete expression and press Enter. |
| Open a step                 | Select it with Up/Down and press Enter.                            |
| Go back                     | Escape: detail to list, list to entry screen.                      |
| Expand / fold a group       | Right / Left in the step list.                                     |
| Read more explanation       | Shift+Tab in the walkthrough.                                      |
| Read the complete text      | T in a walkthrough/browser, or Actions > Read Full Text.           |
| Enable hints                | !h on, then submit the problem.                                    |
| Reveal another hint         | Tab, while the hint walkthrough is open.                           |
| Reopen the last steps       | !!                                                                 |
| Differentiate once          | !d expression                                                      |
| Integrate once              | !i expression                                                      |
| Solve once                  | !s equation                                                        |
| Set the prefix variable     | !v x                                                               |
| Use plain Giac              | !g on its own line.                                                |
| Reset normal entry behavior | !g, then !h off on a separate line.                                |
| Save                        | Ctrl+S.                                                            |
| Identify the build          | !m                                                                 |

## Before you copy an answer

Read the whole expression, its unit, its conditions and its status. A preview can be incomplete. A CAS answer can lack native steps. A partial derivation is not a completed solution.

# Picture guide: Home, Doc and Settings

*Identify the system view before pressing a shortcut*



**These are TI system screens. StepCAS has its own command menu.**

Figure 12. Annotated retained emulator captures. The screens illustrate TI system navigation. They are not StepCAS configuration dialogs.

Home chooses an application or document. Doc manages the open document. Settings and Status contains several different dialogs. Document Settings controls TI mathematical presentation. Follow names rather than memorizing menu numbers, because a different OS or application can show different options.

## Do not choose Restore Factory Defaults to fix a lesson

That item appears in the picture because it exists in the system menu. It is not a step in this manual's troubleshooting procedure.

## Picture guide: two calculator applications

*Learn the visual clues that tell them apart*

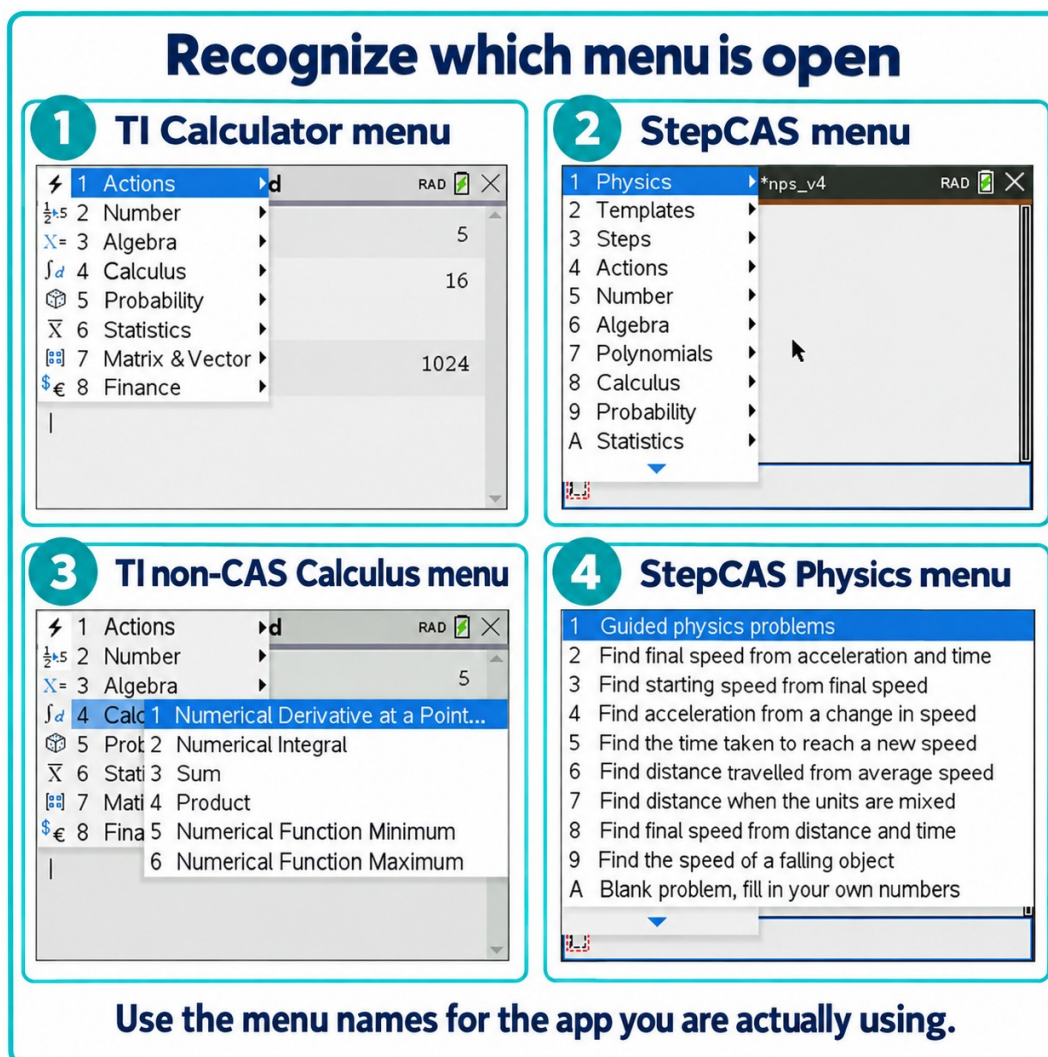


Figure 13. Annotated retained emulator captures comparing TI Calculator and StepCAS. The lower-left panel is the non-CAS TI Calculus menu.

The StepCAS menu contains Physics, Templates and Steps. The TI Calculator menu includes its own Number, Algebra and Calculus commands. A command name from one app is not a guarantee of the same syntax or symbolic capability in the other.

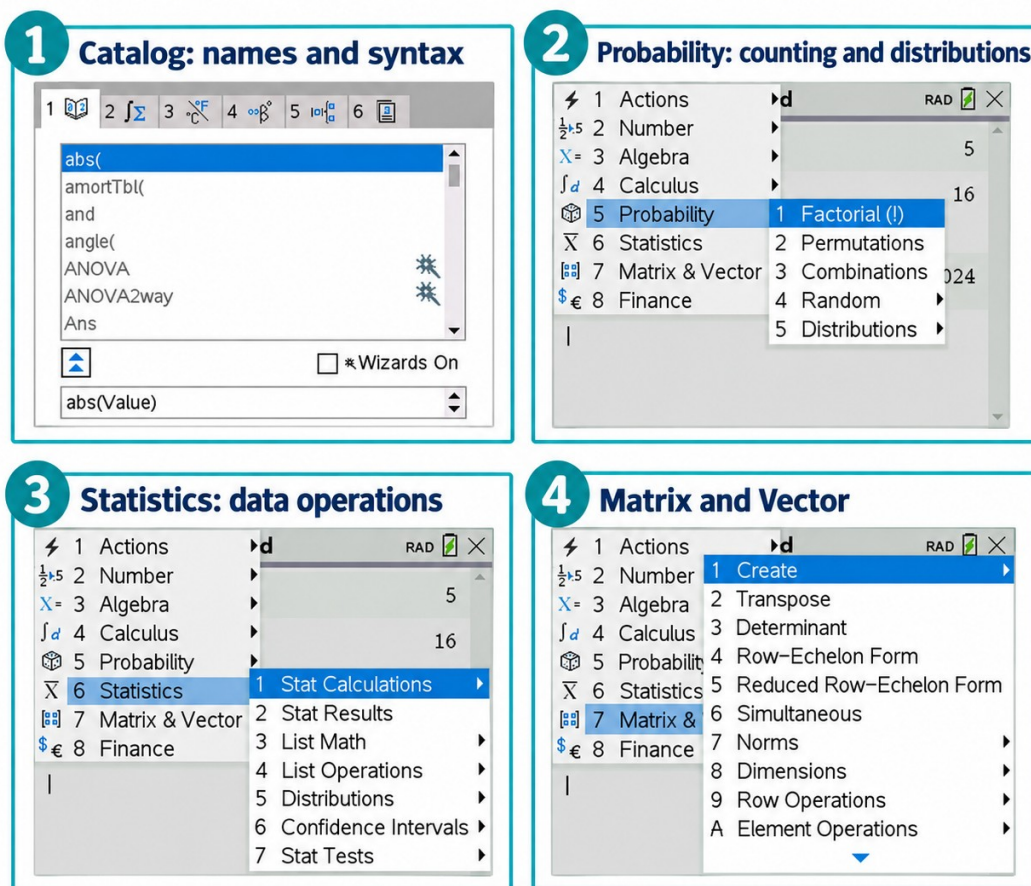
### Your location determines the instructions

If a tutorial tells you to choose Steps > Hint walkthrough, open the StepCAS document first. It will not appear in TI Scratchpad.

## Picture guide: Catalog and native tools

*Read the command name, signature and submenu*

### More tools in TI Calculator



**These built-in tools are separate from StepCAS native walkthroughs.**

Figure 14. Annotated retained TI Calculator captures. These illustrate the built-in catalog and menus, not native StepCAS teaching coverage.

A catalog signature names the arguments a function expects. A submenu marker means there is another list to open. A down marker or scrollbar means the visible rows are not the whole list. Use Enter to open a selection and Escape to return.

#### Read the data question first

A distribution, a test, a statistic and a matrix operation answer different questions. Do not choose a similarly named item simply because it is visible.

## Picture guide: opening the document

*Recognize the launcher and the application screen*

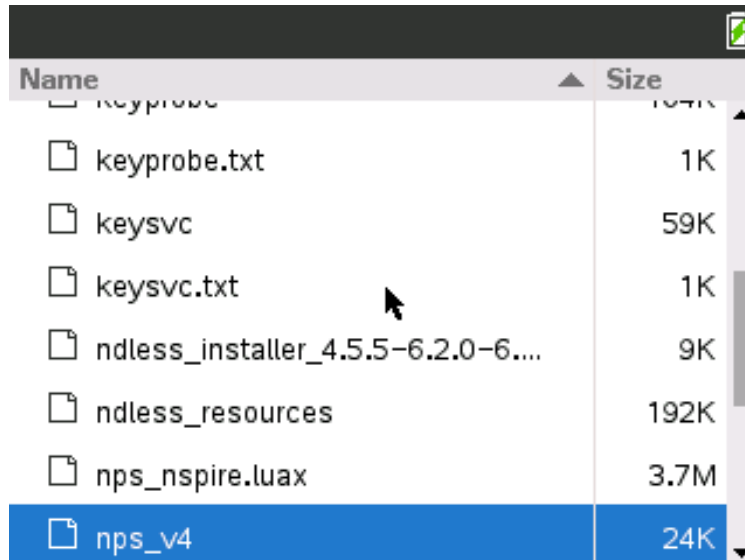


Figure 15. Earlier installation browser capture. nps\_v4 is the selected document. The old neighboring file names and folder arrangement are not the current installation recipe.

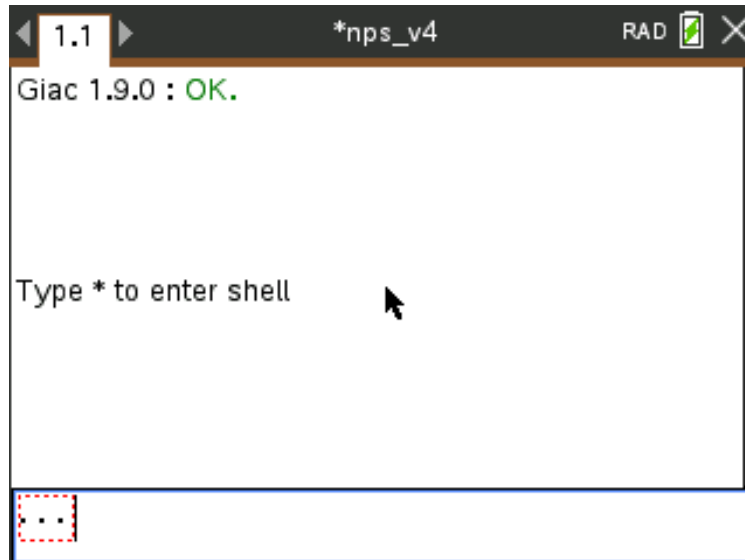


Figure 16. Retained StepCAS entry screen. The bottom editable box is present even before history contains any calculations.

The document is the item you open. The native module is a dependency. If the app opens but reports an integrity or compatibility failure, preserve the message instead of opening a random neighboring file.

## Picture guide: the complete physics description

*Read the footer before running the selected example*

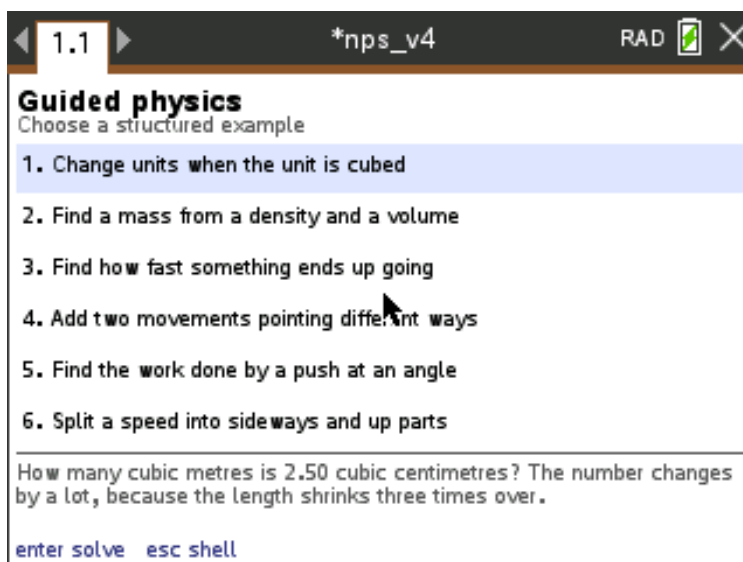


Figure 17. Retained guided browser. The selected row has a pale highlight. The bottom description explains its actual quantities.

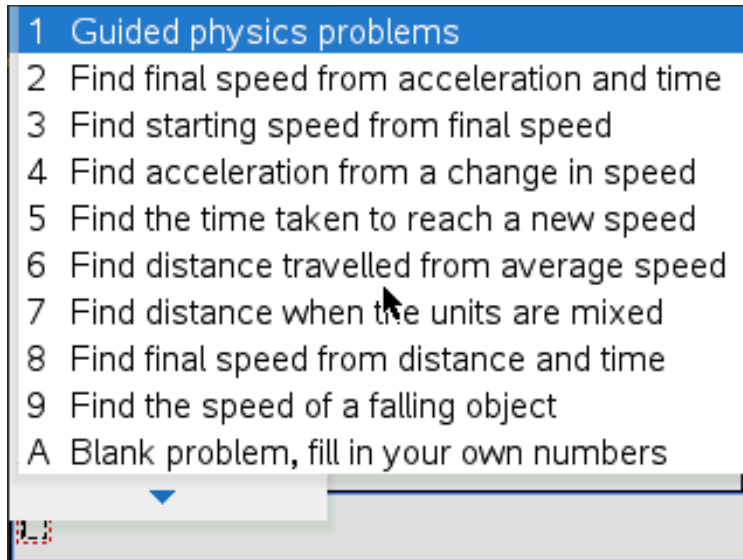


Figure 18. Retained Physics menu. Guided physics problems opens the browser. The other motion entries insert editable kinematics text.

A guided example and a motion template are different entry methods. In the browser, Enter immediately runs the prepared problem. In the template case, inspect and complete the input line before evaluating it.

## Picture guide: answer and step details

*A compact result can have a longer explanation underneath*

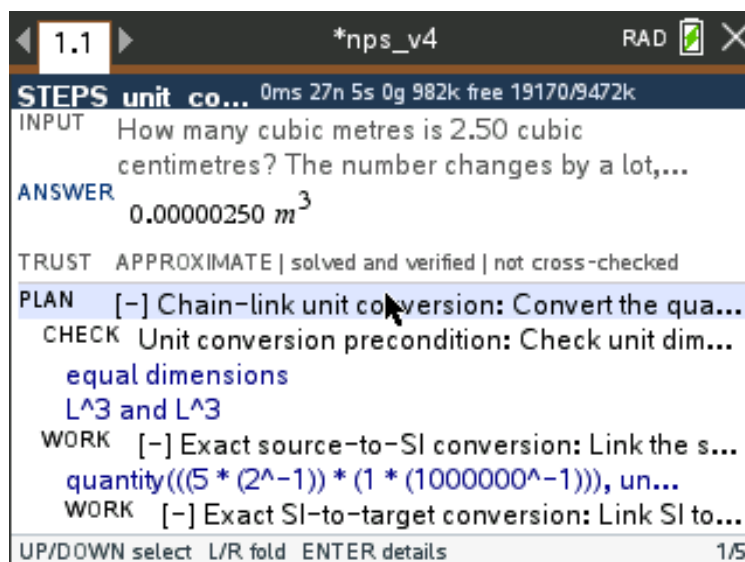


Figure 19. Retained unit-conversion result. The answer is 0.00000250 m<sup>3</sup>. The initial problem text and some rows are shortened by the small screen.

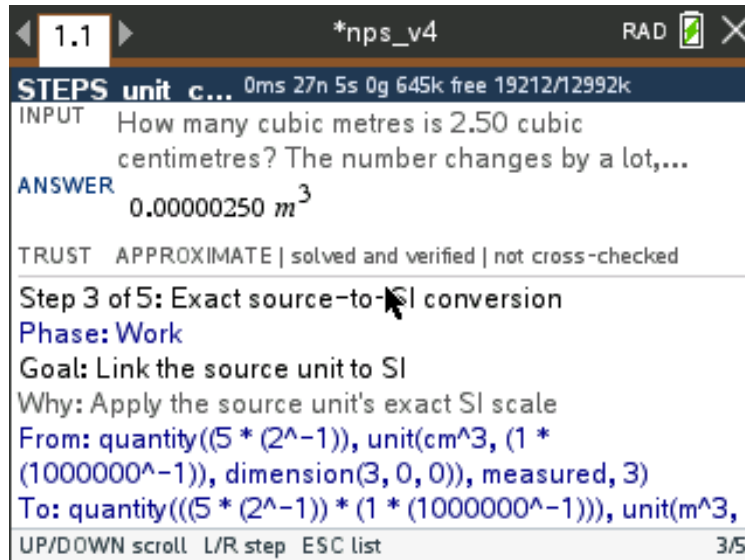


Figure 20. Earlier detail layout for the conversion. This historical build used From and To labels. Current transformation details use Do, Write and Why.

This pair illustrates why you should open details and Full Text. A long internal-looking expression in an old capture is not a string you need to type. Use the current readable instruction fields when your package supplies them.

## Picture guide: fractions, powers and notation

*Pretty display and accepted solver syntax are separate concerns*

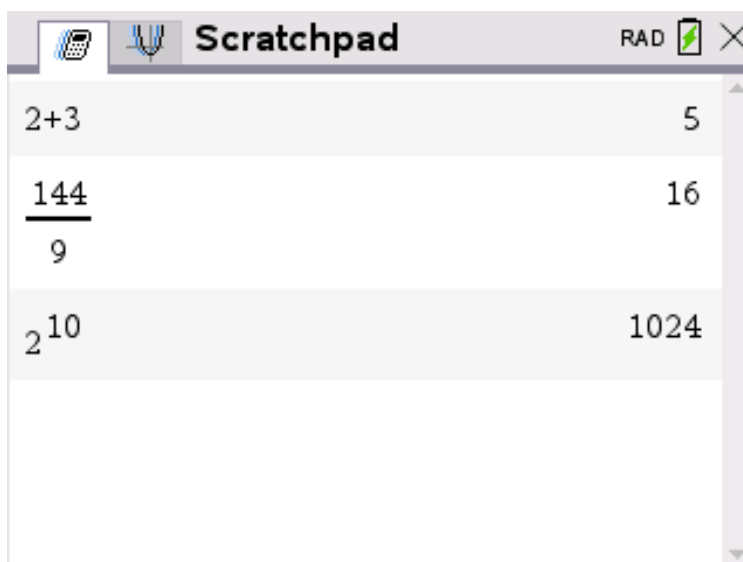


Figure 21. TI Scratchpad renders powers and fractions in two dimensions. The values here are  $2+3=5$ ,  $144/9=16$  and  $2^{10}=1024$ .

| 1.1       |                                                          | *d2_probe                                                                           |                                                                                     | RAD  |  |  |  |
|-----------|----------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------|--|
| frac      | $x+1$                                                    |  |  | 39x54                                                                                     |  |                                                                                       |  |
| power     | $x^{-1}$                                                 |  |  | 28x32                                                                                     |  |                                                                                       |  |
| root      | $\sqrt{x+1}$                                             |  |  | 39x29                                                                                     |  |                                                                                       |  |
| int giac  | $\int x^2 dx$                                            |  |  | 51x32                                                                                     |  |                                                                                       |  |
| int ti    | $\int x^2 dx$                                            |  |  | 51x36                                                                                     |  |                                                                                       |  |
| diff giac | $\frac{d}{dx} x^2$                                       |  |  | 54x32                                                                                     |  |                                                                                       |  |
| vector    | $\begin{bmatrix} 3 \end{bmatrix}$                        |  |  | 25x43                                                                                     |  |                                                                                       |  |
| unit      | $9.0 \cdot 10^3$                                         |  |  | 50x54                                                                                     |  |                                                                                       |  |
| long      | $2 \cdot x^3 - 4 \cdot x^2 + 7 \cdot x - 11 \cdot x + 5$ |  |  | 147x79                                                                                    |  |                                                                                       |  |

Figure 22. Historical renderer test document. It compares a fraction, power, root, Giac integral spelling, TI integral spelling, derivative spelling, vector and unit. Some test rows are clipped by deliberately small boxes.

The renderer can typeset some structures while leaving an unrecognized calculus name as text. StepCAS already adapts integral spelling in walkthrough displays. The editable stacked derivative requested earlier remains an unverified future change. A rendering test document is not the normal StepCAS input screen.

## Picture guide: graph views

*Use the active graph's own hints*

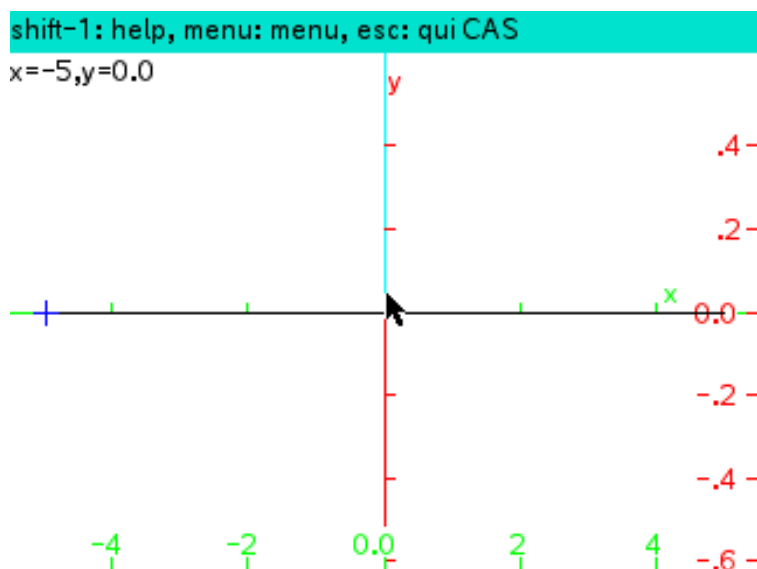


Figure 23. Retained native 2D graphics view. This capture shows axes and a cursor. It does not document a successful parabola plot.

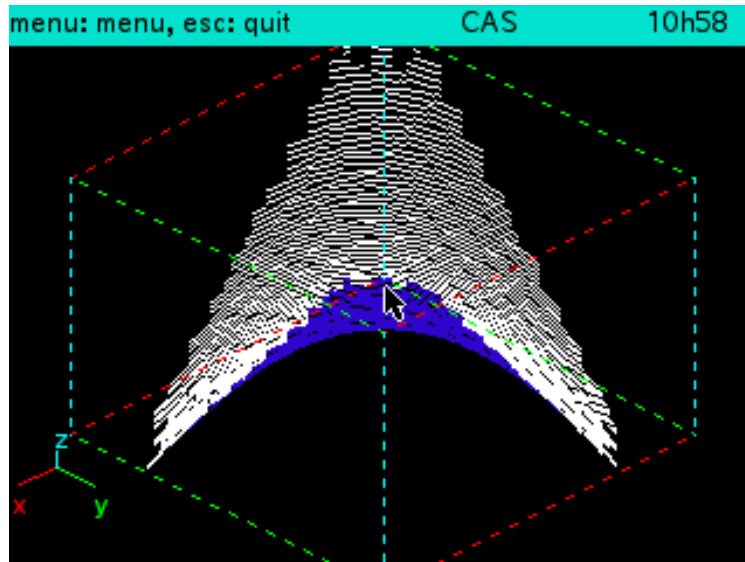


Figure 24. Retained native 3D view. The top strip names its Menu and Escape controls.

Both graphics screens differ from the StepCAS list. If a curve is missing, inspect the graph range and expression. For a first graph, use a simple bounded command from the graph lesson. For advanced 3D commands, use the matching Giac documentation and installed-build checks.

## Menu reference: Physics

### *Current Ki V4 menu inventory*

Guided problems run prepared examples. The remaining entries insert kinematics input that you can edit before Enter.

| Menu label as supplied                      | Meaning and use                    |
|---------------------------------------------|------------------------------------|
| Guided physics problems                     | Runs supplied examples.            |
| Find final speed from acceleration and time | Inserts editable kinematics input. |
| Find starting speed from final speed        | Inserts editable kinematics input. |
| Find acceleration from a change in speed    | Inserts editable kinematics input. |
| Find the time taken to reach a new speed    | Inserts editable kinematics input. |
| Find distance travelled from average speed  | Inserts editable kinematics input. |
| Find distance when the units are mixed      | Inserts editable kinematics input. |
| Find final speed from distance and time     | Inserts editable kinematics input. |
| Find the speed of a falling object          | Inserts editable kinematics input. |
| Blank problem, fill in your own numbers     | Inserts editable kinematics input. |

#### **A label is not a complete command**

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Templates (1)

### *Current Ki V4 menu inventory*

These entries insert expression structures or unit text. Move between native editor slots carefully. They are typing aids rather than promises of solver coverage.

| Menu label as supplied                 | Meaning and use                                         |
|----------------------------------------|---------------------------------------------------------|
| Fraction, one number over another      | Inserts a template or unit text into the visible input. |
| Power, a number raised to another      | Inserts a template or unit text into the visible input. |
| Square root                            | Inserts a template or unit text into the visible input. |
| Cube root, and other roots             | Inserts a template or unit text into the visible input. |
| Derivative, how fast something changes | Inserts a template or unit text into the visible input. |
| Integral, the area under a curve       | Inserts a template or unit text into the visible input. |
| Integral between two limits            | Inserts a template or unit text into the visible input. |
| Equation, two sides that are equal     | Inserts a template or unit text into the visible input. |

#### **A label is not a complete command**

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Templates (2)

### *Current Ki V4 menu inventory*

These entries insert expression structures or unit text. Move between native editor slots carefully. They are typing aids rather than promises of solver coverage.

| Menu label as supplied            | Meaning and use                                         |
|-----------------------------------|---------------------------------------------------------|
| Unit: metres                      | Inserts a template or unit text into the visible input. |
| Unit: seconds                     | Inserts a template or unit text into the visible input. |
| Unit: kilograms                   | Inserts a template or unit text into the visible input. |
| Unit: metres per second           | Inserts a template or unit text into the visible input. |
| Unit: metres per second squared   | Inserts a template or unit text into the visible input. |
| Unit: newtons, for force          | Inserts a template or unit text into the visible input. |
| Unit: joules, for energy and work | Inserts a template or unit text into the visible input. |

#### **A label is not a complete command**

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Steps

### *Current Ki V4 menu inventory*

These choices set full or hint progression. Explanation detail is changed separately with Shift+Tab inside a walkthrough.

| Menu label as supplied       | Meaning and use                             |
|------------------------------|---------------------------------------------|
| Full walkthrough (all steps) | Changes subsequent walkthrough progression. |
| Hint walkthrough (Tab next)  | Changes subsequent walkthrough progression. |

#### **A label is not a complete command**

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Actions

### *Current Ki V4 menu inventory*

These commands affect the shell, display, history or CAS session. Saving variables is separate from saving the document. Restarting CAS is broader than clearing the current line.

| Menu label as supplied                          | Meaning and use                                                           |
|-------------------------------------------------|---------------------------------------------------------------------------|
| Open Shell *                                    | A display action or native shell/CAS command. Read the Actions lesson.    |
| Open Script Editor                              | A display action or native shell/CAS command. Read the Actions lesson.    |
| Read Full Text                                  | A display action or native shell/CAS command. Read the Actions lesson.    |
| Save Variables <code>write("a.tns",0</code>     | Inserts an unfinished file-write call. Complete it carefully. Not Ctrl+S. |
| Read Variables <code>eval(read("a.tns"))</code> | A display action or native shell/CAS command. Read the Actions lesson.    |
| Show/Hide Editor Borders                        | A display action or native shell/CAS command. Read the Actions lesson.    |
| Show/Hide Horizontal Lines                      | A display action or native shell/CAS command. Read the Actions lesson.    |
| Increase Font Size                              | A display action or native shell/CAS command. Read the Actions lesson.    |
| Decrease Font Size                              | A display action or native shell/CAS command. Read the Actions lesson.    |
| Restart CAS <code>restart;</code>               | Session reset. Preserve needed variables first.                           |
| Clear History                                   | Confirmation required. Preserves pending input.                           |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lu/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## 53 / REFERENCE

# Menu reference: Number (1)

## Current Ki V4 menu inventory

Integer and number-theory commands. Some literal-input methods have native walkthroughs. The remaining entries use the backend or remain outside teaching coverage.

| Menu label as supplied                          | Meaning and use                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Convert to Decimal <code>evalf(x[,prec])</code> | Approximate $x$ numerically. The optional precision is a backend setting, not StepCAS Exact/Decimal mode. |
| Factor Integer <code>ifactor(n)</code>          | Write a positive integer as a product of primes. Example: $12 = 2^2 \cdot 3$ .                            |
| Euclidean Quotient <code>iquo(a,b)</code>       | Return the integer quotient in division. Read together with the remainder.                                |
| Euclidean Remainder <code>irem(a,b)</code>      | Return the remainder after integer division. Example: <code>irem(17,5)</code> gives 2.                    |
| Prime Test <code>is_prime(p)</code>             | Test whether an integer is prime. Native status 2 means proven prime and 0 means not prime.               |
| Next Prime <code>nextprime(n)</code>            | Find a prime larger than the supplied integer.                                                            |
| Divisor List <code>idivis(n)</code>             | List the positive integers that divide $n$ exactly.                                                       |
| Extended GCD <code>iegcd(a,b)</code>            | Find the greatest common divisor and coefficients that express it as a combination of the inputs.         |
| Bezout Coefficients <code>iabcuv(a,b,c)</code>  | Find coefficients $u$ and $v$ satisfying $a \cdot u + b \cdot v = c$ , or the polynomial counterpart.     |

### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Number (2)

### *Current Ki V4 menu inventory*

Integer and number-theory commands. Some literal-input methods have native walkthroughs. The remaining entries use the backend or remain outside teaching coverage.

| Menu label as supplied                             | Meaning and use                                                                                   |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Euler Indicatrix <code>euler(n)</code>             | Count integers from 1 through $n$ that are relatively prime to $n$ .                              |
| Chinese Remainder <code>ichrem([a,n],[b,m])</code> | Combine congruences $x=a$ modulo $n$ and $x=b$ modulo $m$ . Compatibility matters.                |
| Power Modulo <code>powmod(a,m,n)</code>            | Evaluate $a^m$ modulo $n$ without first displaying the full power.                                |
| Absolute Value <code>abs(x)</code>                 | Distance from zero for a real number. For a complex number, its magnitude.                        |
| Floor <code>floor(x)</code>                        | Round down toward negative infinity. <code>floor(-1.2)</code> is <code>-2</code> .                |
| Ceiling <code>ceil(x)</code>                       | Round up toward positive infinity. <code>ceil(-1.2)</code> is <code>-1</code> .                   |
| Sign <code>sign(x)</code>                          | Indicate whether a real value is negative, zero or positive.                                      |
| Round <code>round(x[,n])</code>                    | Round a value, optionally controlling decimal places. This changes presentation or approximation. |
| Maximum <code>max(x)</code>                        | Choose the largest value from the supplied values or list.                                        |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Number (3)

### *Current Ki V4 menu inventory*

Integer and number-theory commands. Some literal-input methods have native walkthroughs. The remaining entries use the backend or remain outside teaching coverage.

| Menu label as supplied                                  | Meaning and use                                                               |
|---------------------------------------------------------|-------------------------------------------------------------------------------|
| Minimum <code>min(x)</code>                             | Choose the smallest value from the supplied values or list.                   |
| Interval <code>[inf..sup]</code>                        | Represent a range from a lower endpoint to an upper endpoint.                 |
| Convert to Interval <code>convert(expr,interval)</code> | Request an interval representation. Check the bounds returned by the backend. |
| Real Part <code>re(z)</code>                            | Extract the real component $a$ from $a+bi$ .                                  |
| Imaginary Part <code>im(z)</code>                       | Extract the coefficient $b$ from $a+bi$ .                                     |
| Complex Conjugate <code>conj(z)</code>                  | Change $a+bi$ to $a-bi$ .                                                     |
| Argument <code>arg(z)</code>                            | Return a complex number's angle. Check the active angle convention.           |

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Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Algebra (1)

### *Current Ki V4 menu inventory*

Solve or transform mathematical expressions. Ordinary top-level solve, factor, simplify and the native rearrange command have limited teaching families. The other forms are primarily Giac operations.

| Menu label as supplied                              | Meaning and use                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Solve <code>solve(expr,var)</code>                  | Find values making an equation true. Native teaching coverage is narrower than Giac solving.     |
| Factor <code>factor(expr)</code>                    | Rewrite an expression or polynomial as multiplied factors.                                       |
| Normal Form <code>normal(expr)</code>               | Normalize an algebraic expression, commonly combining rational terms.                            |
| Simplify <code>simplify(expr)</code>                | Request a simpler equivalent form. Conditions and the supported family still matter.             |
| Substitute <code>subst(expr,var,value)</code>       | Replace a variable in an expression with a supplied value or expression.                         |
| Convert <code>convert(expr,...)</code>              | Change representation according to a target option. Read the exact option in the Giac reference. |
| Numerical Solve <code>fsolve(expr,var,guess)</code> | Search for an approximate solution near a guess. One returned root need not be every root.       |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Algebra (2)

### *Current Ki V4 menu inventory*

Solve or transform mathematical expressions. Ordinary top-level solve, factor, simplify and the native rearrange command have limited teaching families. The other forms are primarily Giac operations.

| Menu label as supplied                                    | Meaning and use                                                                              |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Recurrence Solve <code>rsolve(eq,un)</code>               | Solve a relation connecting successive sequence terms. Use the documented sequence notation. |
| Partial Fractions <code>partfrac(expr)</code>             | Split a rational expression into simpler fractional terms.                                   |
| Collect Trig <code>tcollect(expr)</code>                  | Combine trigonometric terms using identities.                                                |
| Expand Trig <code>texpand(expr)</code>                    | Expand trigonometric expressions using identities.                                           |
| Complex Factor <code>cfactor(expr)</code>                 | Factor allowing complex coefficients or roots.                                               |
| Complex Partial Fractions<br><code>cpartfrac(expr)</code> | Decompose a rational expression allowing complex factors.                                    |
| Complex Solve <code>csolve(expr,var)</code>               | Solve with complex solutions permitted.                                                      |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lu/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Polynomials (1)

### *Current Ki V4 menu inventory*

Commands for roots, coefficients, interpolation and polynomial algebra. These advanced menu entries are not all native walkthroughs. Choose the exact command signature from Giac's reference.

| Menu label as supplied                               | Meaning and use                                                                                                              |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Factor <code>factor(P)</code>                        | Rewrite an expression or polynomial as multiplied factors.                                                                   |
| Complex Factor <code>cfactor(P)</code>               | Factor allowing complex coefficients or roots.                                                                               |
| Polynomial Roots <code>proot(P)</code>               | Find polynomial roots numerically. Check repeated roots and numerical precision.                                             |
| Degree <code>degree(P,var)</code>                    | Find the highest polynomial exponent in the selected variable.                                                               |
| Coefficient <code>coeff(P,var,n)</code>              | Extract the coefficient of the selected variable's nth power.                                                                |
| Horner Form <code>horner(P,x)</code>                 | Rewrite a polynomial as nested multiplication for evaluation.                                                                |
| Canonical Form <code>canonical_form(P,var)</code>    | Rewrite a quadratic in completed-square form.                                                                                |
| Coefficients to Polynomial <code>pcoeff(list)</code> | The label is misleading. Giac <code>pcoeff</code> takes a list of roots and returns the monic polynomial's coefficient list. |
| Interpolation <code>lagrange(X,Y)</code>             | Construct a polynomial through paired X and Y values. Match the two lists.                                                   |
| Euclidean Division <code>quorem(A,B)</code>          | Return polynomial quotient and remainder when dividing A by B.                                                               |
| Greatest Common Divisor <code>gcd(A,B)</code>        | Find a greatest common polynomial factor.                                                                                    |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luar/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Polynomials (2)

### *Current Ki V4 menu inventory*

Commands for roots, coefficients, interpolation and polynomial algebra. These advanced menu entries are not all native walkthroughs. Choose the exact command signature from Giac's reference.

| Menu label as supplied                              | Meaning and use                                                                                   |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Extended GCD <code>egcd(A,B,var)</code>             | Find the greatest common divisor and coefficients that express it as a combination of the inputs. |
| Bezout Coefficients <code>abcuv(A,B,C,var)</code>   | Find coefficients $u$ and $v$ satisfying $a*u+b*v=c$ , or the polynomial counterpart.             |
| Polynomial to List <code>symb2poly(P,var)</code>    | Turn a polynomial into its ordered coefficient list.                                              |
| List to Polynomial <code>poly2symb(list,var)</code> | Build a polynomial from an ordered coefficient list in the selected variable.                     |
| Resultant <code>resultant(A,B,var)</code>           | Eliminate a variable using the resultant of two polynomials. This is an advanced algebra tool.    |
| Groebner Basis <code>gbasis(polys,vars)</code>      | Compute a structured basis for a multivariable polynomial system. Variable order matters.         |
| Cyclotomic <code>cyclotomic(n)</code>               | Construct the polynomial for primitive $n$ th roots of unity.                                     |
| Hermite <code>hermite(n)</code>                     | Construct a Hermite polynomial of the requested degree.                                           |
| Chebyshev First <code>tchebyshev1(n)</code>         | Construct a Chebyshev polynomial of the first kind.                                               |
| Chebyshev Second <code>tchebyshev2(n)</code>        | Construct a Chebyshev polynomial of the second kind.                                              |
| Random Polynomial <code>randpoly(n)</code>          | Generate a polynomial using random coefficients. Repeated calls can differ.                       |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

# Menu reference: Calculus

## *Current Ki V4 menu inventory*

Derivative and indefinite-integral commands have limited native lessons. Limits, sums, series and differential equations primarily use Giac. Extra derivative orders and integral bounds can be refused by automatic teaching dispatch. Switch to plain Giac deliberately for those calculations.

| Menu label as supplied                             | Meaning and use                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Derivative <code>diff(expr,var)</code>             | Differentiate the expression with respect to the named variable.                                    |
| Integral <code>int(expr,var)</code>                | Find an antiderivative. A two-argument native request is an indefinite integral.                    |
| Limit <code>limit(expr,var,value)</code>           | Find the value approached as the variable tends to the supplied point. No native limit walkthrough. |
| Sum <code>sum(expr,var,min,max)</code>             | Add expression values over the supplied integer index bounds.                                       |
| Series <code>series(expr,var=value,order)</code>   | Request a local series about the specified point to a chosen order.                                 |
| Differential Equation <code>desolve(eq,x,y)</code> | Solve a differential equation with named independent and dependent variables.                       |

### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## 61 / REFERENCE

# Menu reference: Probability (1)

## Current Ki V4 menu inventory

Factorial, permutation and combination have bounded native counting lessons. Distributions and tests require the right statistical interpretation and generally return backend answers.

| Menu label as supplied                    | Meaning and use                                                                            |
|-------------------------------------------|--------------------------------------------------------------------------------------------|
| Factorial factorial(n)                    | Multiply the integers from 1 through n. 0! is 1.                                           |
| Permutations perm(n,p)                    | Count ordered selections of p objects from n without replacement.                          |
| Combinations comb(n,p)                    | Count selections of p objects from n when order does not matter.                           |
| Random rand()                             | Generate a backend random value. See Giac's variants for a specific range or distribution. |
| Binomial binomial(n,p,k)                  | Probability of exactly k successes in n independent trials with success probability p.     |
| Binomial Cdf binomial_cdf(n,p,x)          | Cumulative binomial probability up to a specified value.                                   |
| Binomial Inverse Cdf binomial_icdf(n,p,y) | Find a binomial quantile from a cumulative probability.                                    |
| Normal normald(m,sigma,x)                 | Normal probability density at x, with mean m and standard deviation sigma.                 |
| Normal Cdf normald_cdf(m,sigma,x)         | Cumulative normal probability up to x.                                                     |

### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Probability (2)

### *Current Ki V4 menu inventory*

Factorial, permutation and combination have bounded native counting lessons. Distributions and tests require the right statistical interpretation and generally return backend answers.

| Menu label as supplied                                     | Meaning and use                                                                                            |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Normal Inverse Cdf<br><code>normald_icdf(m,sigma,y)</code> | Find a normal quantile from a cumulative probability.                                                      |
| Poisson <code>poisson(mu,k)</code>                         | Probability of k events for a Poisson model with mean mu.                                                  |
| Exponential <code>exponentiald(mu,x)</code>                | Exponential density. Confirm whether a parameter is a rate or a scale before entering it.                  |
| Geometric <code>geometric(p,k)</code>                      | Geometric probability for repeated independent trials. Check the trial-count convention.                   |
| Chi Squared <code>chisquared(n,x)</code>                   | Chi-squared density with n degrees of freedom.                                                             |
| Uniform <code>uniformd(a,b,x)</code>                       | Uniform density on the interval from a to b.                                                               |
| Chi Squared Test <code>chisquaret(l1,l2)</code>            | Compare supplied observations with a model using a chi-squared test. Check input list shapes.              |
| Z Test <code>normalt</code>                                | Test a mean using a normal model. Specify data, reference mean, uncertainty and alternative as documented. |
| Student Test <code>studentt</code>                         | Test a mean using a Student t model. Read the required arguments and assumptions first.                    |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Statistics (1)

### *Current Ki V4 menu inventory*

List operations and plot commands. Read the list shape and argument order. A plot or computed statistic is not a native derivation merely because it appears in the menu.

| Menu label as supplied                      | Meaning and use                                                      |
|---------------------------------------------|----------------------------------------------------------------------|
| Sequence <code>seq(expr,var,inf,sup)</code> | Evaluate an expression across an index range.                        |
| Size <code>size(l)</code>                   | Count elements in a list.                                            |
| List to Sequence <code>op(l)</code>         | Expose the elements of a list as a sequence of operands.             |
| Apply Function <code>apply(f,l)</code>      | Apply the same function to each list element.                        |
| Append <code>append(l,expr)</code>          | Add an expression to the end of a list.                              |
| Concatenate <code>concat(l1,l2)</code>      | Join two lists in order.                                             |
| First Element <code>head(l)</code>          | Return the first list element.                                       |
| Tail <code>tail(l)</code>                   | Return the list after removing its first element.                    |
| Sort <code>sort(l[,function])</code>        | Order list elements, optionally with a supplied comparison function. |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Statistics (2)

### *Current Ki V4 menu inventory*

List operations and plot commands. Read the list shape and argument order. A plot or computed statistic is not a native derivation merely because it appears in the menu.

| Menu label as supplied                                      | Meaning and use                                                                                       |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Reverse List <code>revlist(l)</code>                        | Return the list in reverse order.                                                                     |
| Contains <code>contains(l,expr)</code>                      | Check whether the supplied element belongs to a collection. Inspect the backend's returned indicator. |
| Suppress <code>suppress(l,n)</code>                         | Remove the element at index n. Giac list positions are normally counted from zero.                    |
| Remove <code>remove(function,l)</code>                      | Remove all list elements for which the supplied function returns true.                                |
| Bar Plot <code>bar_plot(data_list)</code>                   | Display supplied categories or values as bars.                                                        |
| Histogram <code>histogram(data,xmin,size)</code>            | Group numerical observations into bins. xmin and size control the bin placement.                      |
| Scatter Plot <code>scatterplot(Xlist,Ylist)</code>          | Plot paired values from equally sized X and Y lists.                                                  |
| Regression Plot<br><code>linear_regression_plot(X,Y)</code> | Plot a fitted linear relationship for paired observations. A fit does not establish causation.        |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Matrix & Vector (1)

### *Current Ki V4 menu inventory*

Exact small ref, rref and det calls have current native handlers. Other linear-algebra commands remain backend features. The legacy Bug item is exposed by the menu, but its user-facing purpose is not established here.

| Menu label as supplied                    | Meaning and use                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| Solve Linear System <code>linsolve</code> | Solve simultaneous linear equations. Complete the equation and variable lists.                |
| Determinant <code>det(M)</code>           | Compute a scalar associated with a square matrix. Zero signals singularity.                   |
| Inverse <code>inv(M)</code>               | Find a matrix that multiplies with M to give the identity. Singular matrices have no inverse. |
| Row Reduce <code>rref(M)</code>           | Reduce a matrix to reduced row echelon form, with pivots normalized.                          |
| Row Echelon <code>ref(M)</code>           | Use elimination to produce row echelon form.                                                  |
| Kernel <code>ker(M)</code>                | Find vectors mapped to zero by the matrix.                                                    |
| Image <code>image(M)</code>               | Find a basis for the matrix's column space.                                                   |
| Bug <code>bug(M)</code>                   | Legacy backend item. Purpose not established. Not a beginner exercise.                        |
| Eigenvalues <code>eigenvalues(M)</code>   | Find scalar factors associated with invariant directions of a square matrix.                  |
| Eigenvectors <code>eigenvects(M)</code>   | Find the directions associated with eigenvalues. Read the backend's output arrangement.       |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lu/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Matrix & Vector (2)

### *Current Ki V4 menu inventory*

Exact small ref, rref and det calls have current native handlers. Other linear-algebra commands remain backend features. The legacy Bug item is exposed by the menu, but its user-facing purpose is not established here.

| Menu label as supplied                         | Meaning and use                                                                                |
|------------------------------------------------|------------------------------------------------------------------------------------------------|
| Jordan Form <code>jordan(M)</code>             | Request an advanced canonical form and associated transformation information.                  |
| Matrix Power <code>matpow(M,n)</code>          | Raise a square matrix to a power using matrix multiplication.                                  |
| Dot Product <code>dot(v1,v2)</code>            | Multiply matching vector components and add them.                                              |
| Cross Product <code>cross(v1,v2)</code>        | Compute the oriented three-dimensional vector product.                                         |
| Identity <code>identity(n)</code>              | Create an n-by-n matrix with ones on the main diagonal and zeros elsewhere.                    |
| Build Matrix <code>matrix(n,m,function)</code> | Create a matrix with n rows and m columns using a function for entries.                        |
| Random Matrix <code>randmatrix(n,m,law)</code> | Create a matrix using a selected random law.                                                   |
| Hilbert Matrix <code>hilbert(n)</code>         | Build the special square matrix with reciprocal index-sum entries.                             |
| Vandermonde <code>vandermonde(list)</code>     | Build a matrix of powers from supplied values. Check the documented row and column convention. |
| L1 Norm <code>l1norm(M)</code>                 | Measure size using the backend's L1 norm convention for the supplied vector or matrix.         |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/lua/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Matrix & Vector (3)

### *Current Ki V4 menu inventory*

Exact small ref, rref and det calls have current native handlers. Other linear-algebra commands remain backend features. The legacy Bug item is exposed by the menu, but its user-facing purpose is not established here.

| Menu label as supplied              | Meaning and use                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------|
| L2 Norm l2norm(M)                   | Measure size using the L2 norm. Matrix and vector definitions must be distinguished. |
| Infinity Norm linfnorm(M)           | Measure size using the infinity-norm convention for the supplied object.             |
| Condition Number cond(M,1 2 inf)    | Estimate sensitivity of a matrix problem under the selected norm.                    |
| LU Decomposition lu(M)              | Factor a matrix into triangular pieces, possibly with permutations.                  |
| QR Decomposition qr(M)              | Factor a matrix into orthogonal and triangular pieces.                               |
| Schur Form schur(M)                 | Request an orthogonal or unitary transformation to triangular form.                  |
| Singular Value Decomposition svd(M) | Factor a matrix using singular values and orthogonal or unitary factors.             |
| Singular Values svl(M)              | Return the nonnegative singular values of a matrix.                                  |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

## Menu reference: Plots

### *Current Ki V4 menu inventory*

These entries can open separate native graphics views. A range uses two dots, as in  $x=-5..5$ . The graph's own menu controls the view.

| Menu label as supplied                                 | Meaning and use                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------|
| Function Plot <code>plot(f(x),x=a..b)</code>           | Plot $f(x)$ over the selected $x$ interval.                                   |
| Area Plot <code>plotarea(f(x),x=a..b)</code>           | Display the area associated with a curve over an interval.                    |
| Parametric Plot <code>plotparam([x,y],t=a..b)</code>   | Plot $x$ and $y$ coordinates both controlled by a parameter $t$ .             |
| Polar Plot <code>plotpolar(r(theta),theta=a..b)</code> | Plot radius as a function of angle. Check the angle convention.               |
| Slope Field <code>plotfield(f(t,y),t=t1..t2)</code>    | Draw direction markers associated with a first-order differential equation.   |
| Differential Equation Plot <code>plotode(f,t,y)</code> | Draw a numerical differential-equation solution with the required conditions. |
| Density Plot <code>plotdensity(f(x,y),x,y)</code>      | Use color or intensity to display a two-variable function.                    |
| Contour Plot <code>plotcontour(f(x,y),x,y)</code>      | Draw curves of equal function value.                                          |
| Implicit Plot <code>plotimplicit(f(x,y),x,y)</code>    | Draw a relation described implicitly by $x$ and $y$ .                         |

#### A label is not a complete command

Text such as `expr`, `var`, `n` or `M` describes an argument. Replace it with your own expression, variable, integer or matrix. Optional-argument brackets in labels are documentation, not always literal characters to type.

Inventory source: `nps/luas/nps_v4.lua`. The names above are retained verbatim. Availability in the menu does not establish successful execution on your installed package.

# Command prefixes and diagnostics

*A complete reference for the ! commands*

| Command             | Effect                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------|
| !d expression       | Differentiate using the selected variable. Bare !d selects persistent differentiation.           |
| !i expression       | Integrate. Bare !i selects persistent integration.                                               |
| !s equation         | Solve. Bare !s selects persistent solving.                                                       |
| !k structured input | Solve kinematics. Bare !k selects persistent kinematics.                                         |
| !v name             | Set the variable for prefix and selected-mode requests.                                          |
| !g                  | Select plain Giac and clear the persistent solver mode.                                          |
| !h on / !h hint     | Select hint progression and enable automatic recognition. Does not clear a selected solver mode. |
| !h off / !h full    | Select full progression and enable automatic recognition. Does not clear a selected solver mode. |
| !!                  | Reopen the retained last walkthrough.                                                            |
| !m                  | Display package/backend identity from the capability manifest.                                   |
| !t                  | Run the build-dependent typed-backend diagnostic. This is a maintenance test, not a math lesson. |
| !?                  | An unrecognized prefix requests the built-in prefix help text.                                   |

## Diagnostic results have a limited meaning

!t can be unavailable. Where enabled, it exercises backend comparisons and can write a diagnostic report. A reported comparison count is not a physical screen test or a complete product acceptance result.

# A small mathematical glossary

## *Words used in the explanations*

| Word              | Plain meaning                                                                |
|-------------------|------------------------------------------------------------------------------|
| Expression        | A mathematical object such as $2x+3$ .                                       |
| Equation          | A statement that two expressions are equal.                                  |
| Variable          | A name that can stand for a number or quantity.                              |
| Coefficient       | A factor multiplying a variable or term.                                     |
| Domain            | The values for which an expression or statement applies.                     |
| Assumption        | A condition the model or derivation relies on.                               |
| Equivalent forms  | Different expressions with the same value under the relevant conditions.     |
| Derivative        | A local rate of change.                                                      |
| Antiderivative    | A function whose derivative is the given function.                           |
| Definite integral | An accumulated signed quantity over stated bounds.                           |
| Limit             | A value approached as an input approaches a point or infinity.               |
| Component         | One coordinate of a vector in a specified frame.                             |
| Dimension         | The physical kind of quantity, such as length or time.                       |
| Reference frame   | The axes and viewpoint used to describe a vector or motion.                  |
| Pivot             | A leading nonzero entry used in row reduction.                               |
| CAS               | A computer algebra system that manipulates symbolic expressions.             |
| Verification      | A check supporting a step or result, with the strength stated by the method. |

## Sources and further reading

### *Primary references checked for this edition*

These links provide the original manufacturer and backend references. The manual paraphrases them and checks StepCAS behavior against the local implementation. A desktop Xcas feature is not automatically a supported handheld UI feature.

#### [\[T1\] TI-Nspire CX II Handheld Keys](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_hh\\_gs/gsc\\_cx\\_hh\\_keys.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_hh_gs/gsc_cx_hh_keys.HTML)

#### [\[T2\] TI handheld keyboard shortcuts](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_hh\\_gs/gsc\\_keyboardshortcuts.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_hh_gs/gsc_keyboardshortcuts.HTML)

#### [\[T3\] TI Home screen](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_hh\\_gs/gsc\\_using\\_the\\_home\\_screen.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_hh_gs/gsc_using_the_home_screen.HTML)

#### [\[T4\] TI working with documents](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_docshh/m\\_docshh.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_docshh/m_docshh.HTML)

#### [\[T5\] TI configuring handhelds and status](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_confighh/m\\_confighh.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_confighh/m_confighh.HTML)

#### [\[T6\] TI connecting and transferring files](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_transfiles hh/m\\_transfiles hh.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_transfiles hh/m_transfiles hh.HTML)

#### [\[T7\] TI document settings](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_confighh/hc\\_customizing\\_document\\_settings.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_confighh/hc_customizing_document_settings.HTML)

#### [\[T8\] TI Scratchpad](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_scratchpad/m\\_scratchpad.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_scratchpad/m_scratchpad.HTML)

#### [\[T9\] TI saving documents](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_docshh/wc\\_saving\\_documents.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_docshh/wc_saving_documents.HTML)

## Sources and further reading (continued)

### *Primary references checked for this edition*

These links provide the original manufacturer and backend references. The manual paraphrases them and checks StepCAS behavior against the local implementation. A desktop Xcas feature is not automatically a supported handheld UI feature.

#### [\[T10\] TI recharging the handheld](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_confighh/hc\\_recharging\\_the\\_handheld.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_confighh/hc_recharging_the_handheld.HTML)

#### [\[T11\] TI handheld setup](#)

[https://education.ti.com/html/webhelp/EG\\_TINspire/EN/content/m\\_confighh/hc\\_customizing\\_the\\_handheld\\_setup.HTML](https://education.ti.com/html/webhelp/EG_TINspire/EN/content/m_confighh/hc_customizing_the_handheld_setup.HTML)

#### [\[G0\] Giac/Xcas reference index](#)

[https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd\\_en/cascmd\\_en.html](https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd_en/cascmd_en.html)

#### [\[G1\] Giac univariate limits](#)

[https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd\\_en/cascmd\\_en445.html](https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd_en/cascmd_en445.html)

#### [\[G2\] Giac products](#)

[https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd\\_en/cascmd\\_en146.html](https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd_en/cascmd_en146.html)

#### [\[G3\] Giac unassigning variables](#)

[https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd\\_en/cascmd\\_en070.html](https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/en/cascmd_en/cascmd_en070.html)

#### [\[G4\] Giac graphing documentation](#)

<https://www-fourier.univ-grenoble-alpes.fr/~parisse/giac/doc/fr/casinter/casinter015.html>

#### [\[P1\] TI product photographs](#)

<https://education.ti.com/en/products/calculators/graphing-calculators/ti-nspire-cx-ii-cx-ii-cas>

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# How this edition was checked

## *Keep current implementation and historical illustrations distinct*

The manual was prepared against the working source on 9 September 2026. The repository HEAD identifier was 0961d6e. The working tree contained additional changes. The manual describes the inspected working files rather than claiming a clean released revision.

| Evidence                               | What it establishes                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|
| nps/luau/nps_v4.lua                    | Menu names, keyboard handling, mode changes, hint behavior, the guided fixture list and saving behavior. |
| nps/src/steps/command.cc               | Which ordinary commands are recognized as native walkthrough requests.                                   |
| nps/catalog/families.md                | Declared native mathematical coverage, bounds and qualification limits.                                  |
| nps/README.md and docs/codebase-map.md | Current packaging and UI contracts.                                                                      |
| khi-src/doc/en/cascmd_en.tex           | Local upstream Giac syntax, including limit direction and products.                                      |
| nps/benchmarks/native-ui               | Retained emulator screenshots and historical native UI measurements.                                     |

The numbered native UI captures have a retained scan note identifying an emulated non-CAS CX II on OS 6.4.0.74. The final-prefixed and renderer-test captures are older retained evidence. They are not relabeled as fresh captures of the current package.

### What was not performed

No new solver implementation, package deployment, OS update, runtime installation or physical-handheld acceptance test was performed to create this manual. The prepared examples were checked mathematically and against the declared syntax, not all replayed on the current handheld.

Some remote Giac subsection fetches timed out. Their syntax was checked in the retained upstream documentation in khi-src. The references remain useful links to the original manual.

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## Picture credits and interpretation

### *Where the teaching illustrations came from*

Hardware photos are adapted from Texas Instruments' TI-Nspire CX II/CX II CAS product page [P1]. The manual retains the manufacturer identification. They show product hardware, not a photo session with your personal calculator.

StepCAS and TI screen images come from the project's retained native-ui captures. Annotated plates add titles, callouts, borders and side-by-side comparisons. Their role is to explain locations and screen structure. Small characters can differ in an edited plate, so the unedited source images are retained in the accompanying assets folder.

| Annotated plate                    | Source image names                                                                   |
|------------------------------------|--------------------------------------------------------------------------------------|
| Finding StepCAS                    | ti-home-original.png                                                                 |
| The controls you will use most     | ti-models-original.png                                                               |
| Entering a problem                 | final-shell-opened.png and tool-palette-boxes.png                                    |
| Open and close a group of steps    | final-derivative.png and final-expanded.png                                          |
| Read the answer and its conditions | steps-answer-mathbox.png and final-relative-result.png                               |
| Opening a guided physics example   | physics-menu-labels.png and guided-browser-labels.png                                |
| The calculator system screens      | 01-home.png, 14-doc.png, 15-settings.png and 17-docsettings.png                      |
| Recognize which menu is open       | 04-menu-top.png, tool-palette-boxes.png, 08-calculus.png and physics-menu-labels.png |
| More tools in TI Calculator        | 13-catalog.png, 09-probability.png, 10-statistics.png and 11-matrix.png              |

#### Use the typed examples for exact input

Illustration titles are reading aids. They are not text to type into the calculator. Mathematical examples in this manual are written out separately so the learning steps do not depend on deciphering a small screenshot.